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ECG Simulator

Comprehensive Technical Documentation
Phase 1/2 (Conduction Graph) and Phase 3 (FHN Cable)

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About This Documentation

This document provides exhaustive technical documentation of the ECG Simulator project, designed for four distinct audiences simultaneously:

- **Physicians and medical students:** ECG theory and clinical interpretation with minimal mathematical background required.
- **Bioscientists and physiologists:** Cardiac electrophysiology without requiring programming expertise.
- **Mathematicians and physicists:** Full mathematical rigor and physical modeling details.
- **Software engineers:** Complete implementation details, code structure, and dependencies.

All concepts are explained from first principles. Where appropriate, concepts are stratified so that readers can engage at their level of interest.

Chapter 1

Project Overview

1.1 Purpose and Scope

The ECG Simulator is a Python-based biomedical simulation framework that models the electrical activity of the human heart and generates realistic 12-lead electrocardiograms (ECGs) under normal and pathological conditions. The simulator serves as both a:

1. **Educational tool:** Teaching ECG interpretation and understanding cardiac electrophysiology
2. **Research platform:** Studying the relationship between conduction abnormalities and ECG morphology
3. **Clinical decision support:** Generating reference ECGs for various pathological states

1.2 Key Capabilities

The simulator can model the following clinical scenarios:

- Normal sinus rhythm (NSR) at various heart rates
- Atrial pathologies: atrial fibrillation, atrial flutter, atrial premature contractions
- Atrioventricular (AV) conduction abnormalities: first-, second-, and third-degree AV blocks
- Bundle branch blocks: right bundle branch block (RBBB), left bundle branch block (LBBB)
- Ventricular arrhythmias: premature ventricular contractions (PVCs), ventricular tachycardia

- Accessory pathway conduction: Wolff-Parkinson-White (WPW) syndrome
- Sinus node dysfunction: sinus bradycardia, sinus tachycardia

1.3 Development Phases

The project is organized into three phases:

1.3.1 Phase 1/2: Conduction Graph Engine

The Phase 1/2 implementation uses a **discrete conduction graph** model where:

- The heart is represented as a directed graph of **nodes** (anatomical regions: SA node, RA, LA, AV node, His bundle, ventricular segments)
- **Activation propagates** as breadth-first search (BFS) with physiological conduction velocities
- Each node has fixed **activation delay** from the previous node
- Refractory periods are **hardcoded** for each node
- **ECG dipole contributions** from each node are modeled as Gaussian functions in time

Advantages:

- Extremely fast (microseconds per step)
- All pathologies implemented as parameter overrides
- Clinically calibrated timing and morphology

Limitations:

- Refractory period is an input, not emergent from dynamics
- No continuous state evolution or tissue coupling

1.3.2 Phase 3: FitzHugh-Nagumo Cable Engine

The Phase 3 implementation uses a **1-D monodomain FitzHugh-Nagumo cable** model where:

- The conduction system is represented as a **linear chain of cells** (18 total)

- Each cell solves the **FitzHugh-Nagumo ordinary differential equations** (ODEs)
- Action potential (AP) propagates via **diffusion coupling** between adjacent cells
- Refractory period **emerges** naturally from the slow recovery variable dynamics
- **Multi-beat rhythm** requires no explicit pacemaker programming
- ECG dipoles updated *continuously* as the AP wavefront propagates

Advantages:

- Emergent refractory period (no hardcoding)
- Continuous state evolution (more physiologically faithful)
- Can model effects of rate-dependent APD changes

Limitations:

- Slower than Phase 1/2 ($2.2\times$ real-time on modern hardware)
- Pathologies less developed (manual parameter override)
- Timing requires careful calibration

1.4 User-Selectable Engine

The user can switch between engines via a toolbar combo box. The switch:

- Is seamless (pause current engine, swap, resume)
- Reinitializes the new engine with current parameter values
- Does not modify parameter settings

Chapter 2

Theoretical Foundations: Mathematical and Physical Models

2.1 Cardiac Electrophysiology Primer

2.1.1 The Action Potential

Every cardiac myocyte is electrically excitable. When stimulated, it generates an **action potential** (AP)—a stereotypical time-evolution of transmembrane voltage V that drives heart contraction.

Phases of the Cardiac Action Potential

The AP has five distinct phases (Figure 2.1):

[Diagram - text description in document]

Figure 2.1: Five phases of the cardiac action potential. Phase 0: rapid depolarization (inward Na^+ current). Phase 1: early repolarization (outward K^+ current). Phase 2: plateau (inward Ca^{2+} current balances outward K^+). Phase 3: rapid repolarization. Phase 4: resting state.

Phase 0 (Depolarization): The membrane potential V rapidly depolarizes from resting $V_{\text{rest}} \approx -90$ mV to peak $\approx +30$ mV due to rapid inward Na^+ current. This phase lasts ≈ 1 ms.

Phase 1 (Early Repolarization): Brief outward K^+ current causes V to decline slightly.

Phase 2 (Plateau): Inward Ca^{2+} current balances outward K^+ current, maintaining $V \approx 0$ mV. This phase lasts 100–300 ms depending on cell type and conditions.

Phase 3 (Late Repolarization): Dominant outward K^+ current repolarizes the cell back toward V_{rest} .

Phase 4 (Resting/Diastole): The cell remains at rest potential, ready for the next stimulus.

Refractoriness

During and immediately after an AP, the cell is **refractory**—it cannot be re-excited by an incoming stimulus:

- **Absolute refractory period:** During phases 0–2 and early phase 3, Na^+ channels are inactivated; no re-stimulation possible.
- **Relative refractory period:** Late phase 3, when V is still below threshold. Re-stimulation is possible only with a **supernormal stimulus** (larger than normal).

The duration of refractoriness is roughly proportional to the AP duration (**APD**). Longer APD $\Rightarrow$ longer refractoriness.

2.1.2 Conduction Velocity

Electrical activation spreads from cell to cell via gap junctions (protein channels connecting adjacent myocytes). The speed of this propagation is the **conduction velocity** (CV):

$$\text{CV} \approx 0.5 \sqrt{\frac{D}{\epsilon}} \cdot \frac{\text{dx}}{\text{cell time constant}}$$

where:

- D is the diffusion coefficient (cytoplasmic resistivity)
- ϵ is the excitability (inversely related to refractory period)
- dx is the cell size

Typical values in the heart:

- Atrial myocardium: 0.5–1.0 m/s
- His-Purkinje system: 2–4 m/s (high CV)
- Ventricular myocardium: 0.3–0.5 m/s
- AV node: 0.02–0.05 m/s (slowest, with programmed delay)

2.2 ECG Generation: The Dipole Model

2.2.1 Cardiac Electrical Dipole

At any instant, the heart has a net electrical dipole moment $\mathbf{p}(t)$ pointing from negative to positive charge:

$$\mathbf{p}(t) = \sum_{\text{all myocytes}} q_i(t) \mathbf{r}_i$$

where $q_i(t)$ is the charge state of myocyte i at position $\mathbf{r}_i$.

During the AP:

- **Depolarization phase 0:** Wavefront of depolarization creates a dipole pointing from rest (negative) to depolarized (positive)
- **Plateau phase 2:** Little dipole moment (most of heart is depolarized)
- **Repolarization phase 3:** Wavefront of repolarization creates a dipole pointing from depolarized (negative) to rest (positive)

2.2.2 12-Lead ECG and Lead Vectors

Each of the 12 ECG leads measures the projection of the cardiac dipole onto a fixed spatial direction called the **lead vector**.

The 12 leads comprise:

- **Limb leads:** I, II, III (measure frontal plane: left-right and inferior-superior)
- **Augmented leads:** aVR, aVL, aVF (also frontal plane, normalized)
- **Precordial leads:** V1–V6 (measure transverse plane: anterior-posterior and left-right)

Standard lead vectors (in mV per mV of dipole component) are fixed by convention and embedded in the simulator.

2.2.3 Gaussian Dipole Model

In this simulator, each anatomical node (e.g., RV, LV lateral) contributes to the ECG as a **time-localized dipole**:

$$\mathbf{p}_{\text{node}}(t) = \mathbf{d}_{\text{node}} [g_{\text{depol}}(t - t_{\text{act}}) + A_{\text{repol}} g_{\text{repol}}(t - t_{\text{act}})] \quad (2.1)$$

where:

- $\mathbf{d}_{\text{node}}$ is the **dipole vector** (anatomy-specific, units: mV-m)
- t_{act} is the **activation time** (when the node first crossed threshold)
- $g_{\text{depol}}(t)$ is the **depolarization Gaussian**: $g_d(t) = \exp\left(-\frac{t^2}{2\sigma_d^2}\right) \Theta(t)$ (active for $t > 0$)
- $\sigma_d \approx 12.7$ ms (controls QRS width; Phase 1/2 value)
- $g_{\text{repol}}(t)$ is the **repolarization Gaussian**: $g_r(t) = \exp\left(-\frac{(t-\tau_r)^2}{2\sigma_r^2}\right)$ (peaks at repolarization delay $\tau_r \approx 280$ ms)
- $A_{\text{repol}} \approx -0.3$ (T-wave is opposite polarity to QRS)

The total voltage in lead k is the sum over all nodes:

$$\text{ECG}_k(t) = \sum_{\text{nodes}} \mathbf{L}_k \cdot \mathbf{p}_{\text{node}}(t) \quad (2.2)$$

where $\mathbf{L}_k$ is the **lead vector** for lead k (dimensionless, normalized).

2.3 Model (Phase 3)

2.3.1 Biological Motivation

The model is a two-variable simplified model of the Hodgkin-Huxley equations that captures the essential fast-slow dynamics of cardiac action potentials without explicitly modeling individual ion channels.

2.3.2 The FitzHugh-Nagumo Equations

In normalized units, the 1-D cable equations are:

$$\frac{\partial v}{\partial t} = \frac{D}{\tau} \frac{\partial^2 v}{\partial x^2} + \frac{1}{\tau} \left(v - \frac{v^3}{3} - w + I_{\text{ext}} \right) \quad (2.3)$$

$$\frac{\partial w}{\partial t} = \frac{\varepsilon}{\tau} (v + a - b \cdot w) \quad (2.4)$$

where:

- $v(x, t)$ is the **transmembrane voltage** (normalized: $-1.2 \leq v \leq 2$; unphysiological units for mathematical simplicity)
- $w(x, t)$ is the **slow recovery variable** (models inactivation of inward current and activation of outward current)

- x is the position along the cable
- D is the **diffusion coefficient** (cell-to-cell coupling)
- τ is the **physical time scale** (all times scale inversely with τ)
- ε is the **time scale separation factor** (small $\varepsilon \Rightarrow$ slow w recovery)
- a, b are constants ($a \approx 0.7, b \approx 0.8$)
- I_{ext} is external stimulus current (applied at specific locations during appropriate times)

Fast-Slow Dynamics

The key insight is that v evolves on the **fast timescale** $\sim \tau$, while w evolves on the **slow timescale** $\sim \tau/\varepsilon$:

- **Depolarization (Phase 0)**: v rapidly increases as the depolarizing current dominates
- **Plateau (Phase 2)**: v is high, w slowly increases, gradually opposing the depolarizing current
- **Repolarization (Phase 3)**: Once w is large enough, it repolarizes v back toward rest
- **Refractory period**: w remains large, suppressing v from re-depolarizing. As w slowly decays, v can eventually respond again

Action Potential Duration

The AP duration (APD) is determined by the recovery timescale:

$$\text{APD} \sim \frac{\tau}{\varepsilon \cdot b}$$

For with $\tau = 15$ ms, $\varepsilon = 0.08$, $b = 0.8$:

$$\tau_w = \frac{0.015}{0.08 \times 0.8} = 0.234 \text{ s} = 234 \text{ ms}$$

This recovery time constant drives the refractoriness: ventricular cells take ≈ 234 ms to fully recover. At 70 bpm (cycle length = 857 ms), a cell fires, then 857 ms later is stimulated again when it has recovered $857/234 \approx 91\%$ of the way. Multi-beat ECGs therefore emerge naturally.

2.3.3 Discretization: Monodomain Cable

The cable is discretized into N_{cells} compartments. The Laplacian $\frac{\partial^2 v}{\partial x^2}$ is approximated by finite differences:

$$\left. \frac{\partial^2 v}{\partial x^2} \right|_i \approx \frac{v_{i-1} - 2v_i + v_{i+1}}{\Delta x^2}$$

with no-flux (Neumann) boundary conditions at the ends:

$$v'_0 = v_1 - v_0, \quad v'_{N-1} = v_{N-2} - v_{N-1}$$

2.3.4 Numerical Integration

The discretized system is integrated using explicit Euler with timestep $\Delta t = 0.1$ ms:

$$v_i^{n+1} = v_i^n + \Delta t \left[\frac{D}{\Delta x^2 \tau} (v_{i-1}^n - 2v_i^n + v_{i+1}^n) + \frac{1}{\tau} \left(v_i^n - \frac{(v_i^n)^3}{3} - w_i^n + I_i^n \right) \right] \quad (2.5)$$

$$w_i^{n+1} = w_i^n + \Delta t \frac{\varepsilon}{\tau} (v_i^n + a - b \cdot w_i^n) \quad (2.6)$$

The CFL (Courant-Friedrichs-Lewy) stability condition requires:

$$D \cdot \frac{\Delta t}{\tau \Delta x^2} \leq 0.5$$

For the current values ($D_{\text{max}} = 72$, $\Delta t = 0.1$ ms, $\tau = 15$ ms, $\Delta x = 1$):

$$\text{CFL} = 72 \times \frac{0.0001}{0.015 \times 1^2} = 0.48 < 0.5 \quad \checkmark$$

2.4 Conduction Graph Model (Phase 1/2)

2.4.1 Graph Structure

The heart is modeled as a directed acyclic graph (DAG) with fixed **nodes** (anatomical regions) and fixed **edges** (conduction pathways):

$$\text{Graph} = (V, E) \quad (2.7)$$

where:

- $V = \{\text{SA, RA, LA, AV node, His, LBB, RBB, SEPT, LV, RV, etc.}\}$ (13 nodes in total)
- $E = \{(u, v) : \text{activation can propagate from } u \text{ to } v\}$

2.4.2 Activation Propagation

At each heartbeat, the node fires, triggering a **breadth-first search (BFS)** wave propagation:

1. Queue $\leftarrow$ [SA node]
2. Activation times $t_{\text{act}} \leftarrow$
3. While Queue is not empty:
 - (a) $u \leftarrow$ Dequeue
 - (b) For each neighbor v of u in Graph:
 - i. If v not yet activated:
 - ii. Delay $\Delta t_{\text{edge}} \leftarrow$ conduction delay on edge (u, v)
 - iii. Refractory check: if $(t_{\text{current}} - t_{\text{act}}[v]) \leq$ refractory period of v , skip
 - iv. Otherwise: $t_{\text{act}}[v] \leftarrow t_{\text{act}}[u] + \Delta t_{\text{edge}}$
 - v. Enqueue v

Each node has a **refractory period** (hardcoded, not emergent):

- , atrial nodes: 300 ms (atrial refractoriness)
- node: 300 ms
- Ventricular nodes: 250 ms

The **conduction delay** on each edge is also fixed:

- SA to RA: 0 ms (instantaneous)
- RA to LA: 100 ms
- LA to AV node: 50 ms
- AV node programmed delay: 100 ms (controlled by parameter)
- AV to His: 0 ms
- His to ventricular nodes: node-dependent (0–50 ms)

2.4.3 Pathological Modifications

Pathologies are modeled by **dynamically modifying** the graph structure or conduction parameters:

- **AV Block III°**: Remove all edges from LA to AV node
- **RBBB**: Increase RBB conduction delay by 50 ms
- **Sinus bradycardia**: Increase SA cycle length
- **Atrial fibrillation**: Replace BFS with chaotic re-entrant activation

Chapter 3

Implementation in Python: Core Modules and Functions

This chapter describes the Python implementation of all models in the simulator.

3.1 Project Directory Structure

```
cardiac_sim/+-core/Corecomputationlayer|+-cell_models/Ionicmodels(FHN, Aliev-  
Panfilov, etc.)|+-conduction/Conductiongraph(Phase1/2)|+-tissue/Cablemodel(Phase3)|+  
--ecg/ECGgeneration(leadfields, dipoles)|+-interfaces.pyAbstractbaseclasses|+  
--parameter_model.pyParameterdefinitionsandstorage|--simulation_worker.pyWorkerthreadforsim  
--simulation/High-levelsimulationengines|+-engine.pyAbstractSimulationEngine|+  
--graph_engine.pyPhase1/2implementation|+-cable_engine.pyPhase3implementation|+  
--solver.pyNumericalsolvers(stub)+-gui/PyQt6GUIlayer|+-main_window.pyMainwindow, tool  
--ecg_display.pyECGwaveformplotter|+-parameter_panel.pyParameterstreewidget|+  
--pathology_panel.pyPathologyselector+-plugins/Pathologyplug- insystem|+-  
av_blocks.pyAVblockpathologies|+-bundle_blocks.pyBundlebranchblocks|+-atrial.pyAtrialpatholo  
--ventricular.pyVentricularpathologies|--sinus_rhythms.pySAnodedisorders--main.pyApplicati
```

3.2 Core Interfaces and Abstract Classes

3.2.1 SimulationEngine Abstract Base Class

File: cardiac_sim/core/interfaces.py

```
class SimulationEngine(ABC): @abstractmethod def initialize(self, params: Simula-  
tionParameters) -> None: """Initialize engine with parameters.""" pass  
@abstractmethod def step(self, dt_s : float) -> ECGSample : """Advancebydt_sseconds, returnECG  
@abstractmethod def apply_pathology(self, plugin : AbstractPathologyPlugin) ->  
None : """Applyapathologicalmodification.""" pass
```

```
@abstractmethod def get_state(self) -> SimulationState : """Get current simulation state(HR, time)"""
```

All engines (graph, cable) inherit from this abstract class and must implement all methods.

3.2.2 ECGSample and SimulationState Data Classes

```
@dataclass class ECGSample: timestamp: float Current time in seconds leads: np.ndarray
Shape (12,) with voltages in mV
```

```
@dataclass class SimulationState: time: float heart_rate: float BPM is_running: bool
```

3.3 Implementation: ConductionGraphEngine

File: cardiac_sim/simulation/graph_engine.py (400 lines)

3.3.1 Class Overview

```
class ConductionGraphEngine(SimulationEngine):
    def __init__(self, params: SimulationParameters = None, nodes: dict[str, ECGNode] = None):
    def initialize(self, params: SimulationParameters) -> None: self.params = params self.nodes = build_nodes()
    def build_nodes() -> dict[str, ECGNode]: self.graph = build_physiological_graph() self.reset()
    def step(self, dt_s: float) -> ECGSample: self.time += dt_s
    def check_fire(self) -> bool: Check if SA should fire (based on HR) if self.time >= self.next_fire: self.fire_sa()
    def compute_ecg(self) -> ECGSample: Compute ECG from active beat ecg = self.compute_ecg(self.time) return ECGSample(timestamp = self.time, leads = ecg)
    def fire_sa(self) -> None: BFS propagation from SA node self.beat = self.run_bfs() self.next_fire = self.params.sa_node.cycle_length_ms / 1000.0
    def run_bfs(self) -> BeatRecord: Breadth-first search to compute activation times Returns a BeatRecord
    def compute_ecg(self, t_s: float) -> np.ndarray: Sum Gaussian dipoles from all active nodes...
```

3.3.2 Key Functions

_build_nodes and build_physiological_graph

File: cardiac_sim/core/conduction/graph.py

These functions construct the fixed conduction graph and node parameters:

```
def build_nodes() -> dict[str, ECGNode] : """Return dict of anatomical nodes with ECG dipole parameters
"""SA_NODE": ECGNode(dipole_vector = [-1.0, 0.5, 0.2], mV = m, normalized depolarization duration = 0.010, 100ms)
def build_physiological_graph() -> dict[str, list[str]] : """Return adjacency list of conduction pathway"""
```

_compute_ecg

Sums Gaussian dipoles (Eq. 2.1) over all nodes:

```

def compute_ecg(self, t_s : float) -> np.ndarray : dipole_total = np.zeros(3) 3D dipole vector
if self.beat is not None : for node_name, t_act in self.beat.activation_times.items() : node =
self.nodes[node_name]
    Depolarization Gaussian g_depol = np.exp(-(t_s - t_act) ** 2 / (2 * node.depol_duration *
** 2)) if t_s < t_act : g_depol = 0
    Repolarization Gaussian (peaks at t_act + repol_delay) t_repol_peak = t_act + node.repol_delay g_repol =
np.exp(-(t_s - t_repol_peak) ** 2 / (2 * node.repol_duration ** 2))
    Total dipole contribution dipole = node.dipole_vector * (g_depol - node.repol_amplitude *
g_repol) dipole_total += dipole
    Project onto 12 lead vectors ecg_12lead = self.lead_field.project(dipole_total).shape(12,) return ecg_12lead

```

3.4 Implementation: ConductionCableEngine

File: cardiac_sim/simulation/cable_engine.py (450 lines)

3.4.1 Class Overview

```

class ConductionCableEngine(SimulationEngine):
    def __init__(self, self.cable: ConductionCable1D = None, DFHN_cable: self.nodes =
    def initialize(self, params: SimulationParameters) -> None: self.params = params self.cable =
    ConductionCable1D(params) self.nodes = build_cable_nodes() adjusted Gaussians self.time =
    0.0
    def step(self, dt_s : float) -> ECGSample : self.time += dt_s
    SA pacemaker if self.time >= self.next_safire : self.fire_sa()
    Always advance cable (to keep internal clock synchronized) newly = self.cable.advance(dt_s)
    If beat in progress, collect nodes as they activate if self.beat_in_progress : for node, t_act in newly.items()
    if node not in self.current_beat_nodes : self.current_beat_nodes[node] = t_act KEY FIX : update active beat time
    self.active_beats[-1].activation_times[node] = t_act
    Commit beat when done ... (beat completion logic)
    Compute ECG from all active beats ecg = self.compute_ecg(self.time) return ECGSample(timestamp=
    self.time, leads = ecg)

```

3.4.2 Cable Propagation: ConductionCable1D

File: cardiac_sim/core/tissue/cable_1d.py (500 lines)

Class Overview and Fixed Parameters

Fixed FHN model parameters (Phase 3-D calibrated) $A = 0.700$ FHN parameter $a_B = 0.800$ FHN parameter b_E $PS = 0.080$ recovery rate T $AU = 0.015$ timescale (15ms for correct PR/QR stimulation) D $T_{INT} = 0.0001$ internal micro-step (0.1ms)

Cable structure ${}_NA = 8atrialcells(SA+7){}_NV = 10ventricularcells(His + LBB(3) + LV(6))$

Diffusion coefficients ${}_DATR = np.array([0.0, 2.88, 2.88, 2.88, 2.88, 2.88, 2.88, 2.88]){}_DVEN = np.array([0.0, 72.0, 72.0, 72.0, 3.0, 3.0, 3.0, 3.0, 3.0, 3.0])$

Stimulus ${}_STIM_AMP = 1.5suprathreshholdcurrent{}_STIM_DUR = 0.01010ms$

```
class ConductionCable1D: def init(self, params: SimulationParameters): self.sa: np.ndarray = np.empty(({}_NA, 2))(cells, [v, w]) self.sv: np.ndarray = np.empty(({}_NV, 2))

def new_beat(self, sa_fire_time : float, params : SimulationParameters) -> None :
    """Reset for a new beat at sa_fire_time.""" self.t = sa_fire_time self.s_time_end_a = sa_fire_time + {}_STIM_DUR
    self.av_pending_a_t = math.inf Reset activation trackers self.ta_ct_a[:] = math.inf self.ta_ct_v[:] = math.inf
    self.ta_ct_a[0] = sa_fire_time

def advance(self, dt_s : float) -> dict[str, float] : """Integrate by dt_s seconds, return newly activated nodes
    max(1, round(dt_s / {}_DT_INT)) dt_micro = dt_s / n_micro newly : dict[str, float] = {}
    for i in range(n_micro) : self.t += dt_micro self.s_step_atrial(dt_micro) self.s_step_ventricular(dt_micro)
    Check for threshold crossings for node, (i0, i1) in {}_ATTRS_EGS.items() : (threshold_detection_logic)...
    for node, (i0, i1) in {}_VEN_EGS.items() : (threshold_detection_logic)...
    return newly

def s_step_atrial(self, dt : float) -> None : """One FHN Euler step on atrial cable.""" V =
    np.ascontiguousarray(self.sa[:, 0]) W = np.ascontiguousarray(self.sa[:, 1]) I = np.zeros({}_NA) if self.t < self.s_time_end_a :
    I[0] = {}_STIM_AMP V_new, W_new = f_hn_cable_step(V, W, {}_DATR, I, dt, {}_TAU_A, {}_TAU_B, {}_EPS) self.sa[:, 0] = V_new
    self.sa[:, 1] = W_new
```

Numerical Integration: `_fhn_cable_step`

File: `cardiac_sim/core/tissue/cable_1d.py`

```
def f_hn_cable_step(V, W, D, I_ext, dt, tau_s, a, b, eps) : """Single explicit Euler step for FHN cable.
    Implements equations (1) and (2) with finite-difference Laplacian. """ Compute Laplacian with no-flux BCs
    lap = np.empty_like(V) lap[1 : -1] = V[1 : -2] - 2.0 * V[1 : -1] + V[2 : ] lap[0] = V[1] - V[0] lap[-1] = V[-2] - V[-1]
    inv_tau = 1.0 / tau_s dv = (D * lap + V - V * 3 / 3.0 - W + I_ext) * inv_tau dw = eps * (V + a - b * W) * inv_tau
    return V + dt * dv, W + dt * dw
```

This function is Numba-JIT compilable (if compatible NumPy version) for $10\text{-}20\times$ speedup.

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commentstyle=gray, stringstyle=red, showstringspaces=false, frame=single,  
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Chapter 1

GUI Description: Complete Reference

File: `cardiac_sim/gui/main_window.py`

1.1 Main Window Architecture

The application window has three main regions:

[Diagram - text description in document]

Figure 1.1: Main window layout. Toolbar at top; left sidebar with parameter/pathology panels; central area for ECG display.

1.1.1 Toolbar

Located at the top, the toolbar provides:

1. **Run/Pause button:** Toggle simulation on/off
 - Enabled: simulation running, ECG updating continuously
 - Disabled: simulation paused, ECG frozen
 - Default: running
2. **Reset button:** Clear ECG history and restart simulation from $t = 0$
 - Clears all buffers
 - Reinitializes engine
 - Restarts timer
3. **Engine selector combo box:** Choose between and

- Option 1: “Phase 1/2 – Conduction Graph” (default)
 - Option 2: “Phase 3 – FHN Cable”
 - Selecting triggers engine swap with no parameter loss
 - Implementation: `_on_engine_changed` in `main_window.py`
4. **Heart rate display:** Reads parameters from engine and shows current HR
- Format: “HR: XX bpm”
 - Updated every GUI frame (typically 30–60 Hz)

1.2 Left Sidebar: Parameter Panel

File: `cardiac_sim/gui/parameter_panel.py`

1.2.1 Overview

The left sidebar contains a hierarchical parameter tree. Each parameter is a **named value** with:

- **Name:** descriptive label
- **Type:** numeric (int, float), boolean, or choice
- **Default value:** physiological normal
- **Minimum/maximum:** physiologically plausible range
- **Units:** seconds, milliseconds, dimensionless, etc.
- **Current value:** user-adjustable

The tree is organized into categories (sections) for clarity:

```
blackParameters:
+- SA Node
| +- Cycle Length (ms)
| +- Ectopic Trigger Rate (beats/min)
| - [other SA parameters]
+- AV Node
| +- AV Delay (ms)
| - [others]
+- Refractory Periods
```

```

| +- Atrial (ms)
| +- Ventricular (ms)
| - [others]
+- Phase 3 Model (if Phase 3 selected)
| +- Cable Diffusion Atrial (units)
| +- Cable Diffusion Ventricular (units)
| - Time Scale Tau (ms)
- [other categories]

```

1.2.2 Complete Parameter List with Ranges and Meanings

	l	c	c	p3cm	
Parameter Name	Min	Default	Max	Meaning	
SA Node Section					
Cycle Length (ms)	300	857	2000	Time between SA fires; inverse of heart rate	
Ectopic Trigger Rate (bpm)	0	0	200	Ectopic atrial pacemaker rate if enabled	
Atrial Section					
RA Conduction Delay (ms)	0	50	200	Extra delay in right atrium	
LA Conduction Delay (ms)	0	50	200	Extra delay in left atrium	
RA-LA Conduction Delay (ms)	0	100	300	RA to LA propagation time	
AV Node Section					
AV Delay (ms)	50	100	300	Programmed AV node delay	
AV Conductance	0	1.0	1.0	0 = complete block; 1 = normal conduction	
AV Refractory Period (ms)	200	300	400	Minimum interval before re-excitation	
Ventricular Section					
RV Activation Delay (ms)	0	20	100	Extra delay in right ventricle	
LV Activation Delay (ms)	0	30	100	Extra delay in left ventricle	
SEPT Activation Delay (ms)	0	10	50	Septal activation delay	
Septal Conductance	0	1.0	1.0	0 = block septal conduction	
RBB Conductance	0	1.0	1.0	0 = complete RBBB; 1 = normal	
LBB Conductance	0	1.0	1.0	0 = complete LBBB; 1 = normal	
Ventricular Refractory Period (ms)	200	250	350	Ventricular refractoriness	
Phase 3 Model (if enabled)					
Cable Time Scale τ (ms)	5	15	30	FHN time constant; affects overall speed	
Atrial Diffusion Coefficient	0.5	2.88	10.0	Atrial cell-to-cell coupling	
Ventricular Diffusion LBB	50	72	150	His-LBB-LV diffusion (fast pathway)	
Ventricular Diffusion LV	1	3.0	20	LV myocardial diffusion (slower)	
ECG Display					
Display Window (s)	2	10	30	Horizontal axis span	
Amplitude Scale (mV/div)	0.1	0.5	2.0	Vertical scale; standard ECG = 0.5 mV/div	

1.2.3 Parameter Data Model

File: cardiac_sim/core/parameter_model.py

```

black@dataclass
    """Central holder of all tunable parameters."""

    # SA node
    sa_node: SANodeParams

```

clas

```
# Conduction delays
atrial: AtrialParams
av_node: AVNodeParams
ventricular: VentricularParams

# Refractory periods (Phase 1/2)
refractoriness: RefractoryParams

# Phase 3 parameters
cable_model: CableModelParams
```

```
# ECG display
ecg_display: ECGDisplayParams
```

Each parameter group is a dataclass with validation:

```
black@dataclass
class AVNodeParams:
    av_delay_ms: float = 100.0    # Normalized to 100 ms
    conductance: float = 1.0      # 0-1 scale: 0=block, 1=normal
    refractory_ms: float = 300.0

    def __post_init__(self):
        """Validate ranges."""
        assert 50 <= self.av_delay_ms <= 300, "AV delay out of range"
        assert 0 <= self.conductance <= 1, "Conductance must be 0-1"
```

1.3 Left Sidebar: Pathology Panel

File: cardiac_sim/gui/pathology_panel.py

1.3.1 Overview

The Pathology Panel is a **list of checkboxes**, each representing a pathological condition. When checked, the condition is applied to the current engine.

1.3.2 Available Pathologies

Pathology Name		Description and Implementation
		AV Blocks

First-Degree AV Block	Increase AV delay by 50 ms (total 150 ms). PR interval prolonged; QRS unchanged.
Second-Degree AV Block (Mobitz I)	Progressively increase AV delay; eventually a beat is dropped. Not implemented in current version.
Second-Degree AV Block (Mobitz II)	Randomly drop 1 in N beats from AV conduction.
Third-Degree AV Block (Complete)	Set AV conductance = 0. Ventricular escape rhythm emerges (junctional or idioventricular).
Bundle Branch Blocks	
Right Bundle Branch Block (RBBB)	Set RBB conductance = 0. RV is activated late via collateral LV conduction; QRS widened, characteristic terminal R' in V1.
Left Bundle Branch Block (LBBB)	Set LBB conductance = 0. LV is activated late via collateral RV conduction; QRS widened, broad notched deflection in lateral leads.
Atrial Pathologies	
Atrial Fibrillation	Replace orderly BFS with chaotic re-entrant activation. Atrial rate 300–600 bpm; erratic baseline; absent P waves.
Atrial Flutter	Regular re-entrant circuit; atrial rate 250–350 bpm; regular “sawtooth” pattern visible in II, III, aVF.
Sinus Bradycardia	Increase SA cycle length to 1200 ms (50 bpm). Normal QRS and T morphology; just slower rate.
Sinus Tachycardia	Decrease SA cycle length to 400 ms (150 bpm). Normal morphology; faster rate.
Ventricular Pathologies	
Premature Ventricular Contraction (PVC)	Trigger ectopic ventricular activation at preset time (usually 600 ms post-SA). Produces wide QRS; T-wave opposite polarity.
Ventricular Tachycardia	Repetitive PVCs at rate 120–250 bpm. Rapid wide complexes; may degenerate to VF.
Ventricular Fibrillation	Chaotic ventricular activation; no organized QRS; baseline appears “fine” or “coarse” fibrillary waves.
Accessory Pathways	
Wolff-Parkinson-White (WPW)	Add accessory pathway bypassing AV node; early ventricular pre-excitation creates short PR and delta wave.
Atrioventricular Reentrant Tachycardia (AVNRT)	Dual AV node pathways; setup conditions for re-entrant circuit; produces regular narrow-complex tachycardia.

1.3.3 Pathology Plugin System

File: `cardiac_sim/pathology/__init__.py` and `cardiac_sim/plugins/`

All pathologies are implemented as **plugins**—modular classes inheriting from an abstract base:

```

blackclass AbstractPathologyPlugin(ABC):
    """Base class for all pathologies."""

    @property
    @abstractmethod
    def name(self) -> str:
        """Pathology name for UI."""

```

```
pass
```

```
@abstractmethod
```

```
def apply(self, engine: SimulationEngine, params: SimulationParameters) -> None:
    """Apply pathological modifications to engine and/or parameters."""
    pass
```

```
@abstractmethod
```

```
def unapply(self, engine: SimulationEngine, params: SimulationParameters) -> None:
    """Restore normal state."""
    pass
```

Example implementation (RBBB):

```
black# File: cardiac_sim/plugins/bundle_blocks.py
```

```
class RBBBPathology(AbstractPathologyPlugin):
```

```
    @property
```

```
    def name(self) -> str:
        return "Right Bundle Branch Block"
```

```
    def apply(self, engine: SimulationEngine, params: SimulationParameters) -> None:
        # Store original for later restoration
        self._original_rbb_conductance = params.ventricular.rbb_conductance
```

```
        # Apply block
        params.ventricular.rbb_conductance = 0.0
```

```
        # Some engines may need explicit reinitialization
        engine.apply_pathology(self)
```

```
    def unapply(self, engine: SimulationEngine, params: SimulationParameters) -> None:
        # Restore
        params.ventricular.rbb_conductance = self._original_rbb_conductance
        engine.apply_pathology(None) # Clear pathology
```

1.3.4 Pathology GUI Interaction

When the user checks a pathology checkbox:

1. GUI calls `apply` on the plugin

2. Plugin modifies parameters and/or engine state
3. Engine is reinitialized with new parameters (seamlessly)
4. ECG display updates to show new morphology

When unchecked:

1. GUI calls `unapply`
2. Original parameters restored
3. Engine reinitialized
4. ECG returns to normal

1.4 Central Display: ECG Plotter

File: `cardiac_sim/gui/ecg_display.py`

1.4.1 Overview

The ECG display is a custom PyQt6 widget using **pyqtgraph** for high-performance waveform rendering.

```
blackclass ECGDisplayWidget(pg.GraphicsLayoutWidget):

    def __init__(self, parent=None):
        super().__init__(parent)

        # Create 12 plot items (one per lead)
        self._plots = {}
        for lead_name in LEAD_NAMES: # ['I', 'II', ..., 'V6']
            plot = self.addPlot(title=lead_name, axisItems={'bottom': TimeAxis()})
            curve = plot.plot(pen='blue')
            self._plots[lead_name] = {'plot': plot, 'curve': curve, 'data': []}

    def update_ecg(self, ecg_sample: ECGSample) -> None:
        """Append new sample and refresh display."""
        for i, lead_name in enumerate(LEAD_NAMES):
            self._plots[lead_name]['data'].append(ecg_sample.leads[i])

        # Keep only last N samples (for time window)
```



```

        if len(self._plots[lead_name]['data']) > self.max_samples:
            self._plots[lead_name]['data'].pop(0)

        # Update plot
        self._plots[lead_name]['curve'].setData(
            self._plots[lead_name]['data'],
            pen=pg.mkPen('blue', width=2)
        )

```

1.4.2 Rendering Parameters

- **Display window:** Default 10 seconds; configurable in toolbar
- **Amplitude scale:** Default 0.5 mV/div (clinical standard); configurable
- **Refresh rate:** 30–60 Hz (user-transparent)
- **Grid:** Major gridlines every 200 ms (horizontal) and 0.5 mV (vertical)
- **Color:** Blue curves, black background

1.4.3 Performance Considerations

To maintain real-time performance:

- Data kept in circular buffers (avoid continuous reallocation)
- Rendering occurs on separate Qt thread (GUI remains responsive)
- Downsampling applied if necessary (though typically unnecessary)

1.5 Simulation Worker Thread

File: `cardiac_sim/core/simulation_worker.py`

1.5.1 Purpose

The simulation engine runs on a separate thread to prevent GUI freezing. The `SimulationWorker` class encapsulates this threading:

```

blackclass SimulationWorker(QThread):

    sample_generated = pyqtSignal(ECGSample) # Emitted when ECG sample ready

```

```
error = pyqtSignal(str)

def __init__(self):
    super().__init__()
    self._engine: SimulationEngine = ConductionGraphEngine()
    self._is_running: bool = False
    self._params: SimulationParameters = SimulationParameters()

def run(self) -> None:
    """Main simulation loop (runs on worker thread)."""
    dt = 0.002 # 2 ms per frame (500 Hz simulation rate)

    while self._is_running:
        sample = self._engine.step(dt)
        self.sample_generated.emit(sample)
        # Throttle to ~500 Hz
        time.sleep(dt / 2) # Rough throttling

def set_engine(self, new_engine: SimulationEngine, params: SimulationParameters) -> None:
    """Swap engines (must be called when NOT running)."""
    if self._is_running:
        raise RuntimeError("Cannot swap engines while running")

    self._engine = new_engine
    self._params = params
    self._engine.initialize(params)
```

1.6 Data Flow Diagram

[Diagram - text description in document]

Figure 1.2: Data flow from UI components through engines to ECG display.

Chapter 2

Detailed Implementation: Cable Engine and Cable Model

2.1 ConductionCableEngine: Complete Details

File: cardiac_sim/simulation/cable_engine.py (450 lines)

2.1.1 Overview

The `ConductionCableEngine` wraps a `ConductionCable1D` numerical model and manages beat state, ECG computation, and multi-beat rhythm.

2.1.2 Key Attributes

```
blackclass ConductionCableEngine(SimulationEngine):
    self._cable: ConductionCable1D = None
    self._params: SimulationParameters = None
    self._nodes: dict[str, ECGNode] = {}
    self._forward_model: LeadFieldModel = None

    # Beat management
    self._time: float = 0.0
    self._next_sa_fire: float = 0.0
    self._active_beats: list[BeatRecord] = [] # Up to 3 concurrent beats
    self._beat_in_progress: bool = False
    self._current_beat_nodes: dict[str, float] = {}

    # Performance tracking
    self._step_count: int = 0
```

2.1.3 Initialization: initialize

```
blackdef initialize(self, params: SimulationParameters) -> None:
    self._params = params
    self._cable = ConductionCable1D(params)
    self._nodes = _build_cable_nodes(params) # Adjusted Gaussians
    self._forward_model = LeadFieldModel()
    self._time = 0.0
    self._next_sa_fire = 0.0
    self._active_beats.clear()
```

2.1.4 Main Simulation Loop: step

This is the **critical function** that drives each simulation step:

```
blackdef step(self, dt_s: float) -> ECGSample:

    KEY PRINCIPLE: Always advance cable (never skip based on beat state).
    This ensures internal cable clock is synchronized with wall clock.
    """
    self._time += dt_s

    # --- SA pacemaker -----
    if self._time >= self._next_sa_fire:
        self._fire_sa()

    # --- Always integrate cable (CRITICAL) -----
    newly_activated = self._cable.advance(dt_s)

    # --- Collect newly activated nodes -----
    if self._beat_in_progress:
        for node_name, t_act in newly_activated.items():
            if node_name not in self._current_beat_nodes:
                self._current_beat_nodes[node_name] = t_act

        # KEY FIX: update active beat immediately
        # so ECG renders QRS at activation time
        if self._active_beats:
            self._active_beats[-1].activation_times[node_name] = t_act
```

```

# --- Check beat completion -----
self._check_beat_completion()

# --- Compute ECG -----
ecg = self._compute_ecg(self._time)

return ECGSample(timestamp=self._time, leads=ecg)

```

2.1.5 SA Pacemaker: `_fire_sa`

```

blackdef _fire_sa(self) -> None:
    cycle_len = self._params.sa_node.cycle_length_ms / 1000.0
    self._next_sa_fire += cycle_len

    # Start new beat
    beat_record = BeatRecord(
        beat_number=len(self._active_beats),
        sa_fire_time=self._time,
        activation_times={}, # Will populate as nodes activate
        qrs_start=None,
        qrs_end=None
    )
    self._active_beats.append(beat_record)
    self._beat_in_progress = True
    self._current_beat_nodes.clear()

    # Trigger cable to begin new beat
    self._cable.new_beat(self._time, self._params)

```

2.1.6 Beat Completion: `_check_beat_completion`

A beat is considered complete when:

1. All ventricular nodes have been activated, OR
2. Time exceeds $0.9 \times$ cycle length (adaptive timeout)

```

blackdef _check_beat_completion(self) -> None:
    """Commit beat when all nodes activated or timeout reached."""
    if not self._beat_in_progress or not self._active_beats:
        return

```

```

beat = self._active_beats[-1]
cycle_len = self._params.sa_node.cycle_length_ms / 1000.0

# All ventricular nodes activated?
ven_nodes = set(VENTRICULAR_NODES) # pre-defined
activated = set(beat.activation_times.keys())
ven_complete = ven_nodes.issubset(activated)

# Timeout check
time_limit = beat.sa_fire_time + 0.9 * cycle_len
timeout_reached = self._time >= time_limit

if ven_complete or timeout_reached:
    beat.is_committed = True
    self._beat_in_progress = False

# Store QRS timing for diagnostics
if beat.activation_times:
    times_v = [
        t for n, t in beat.activation_times.items()
        if n in VENTRICULAR_NODES
    ]
    if times_v:
        beat.qrs_start = min(times_v)
        beat.qrs_end = max(times_v)

```

2.1.7 ECG Computation: `_compute_ecg`

```

blackdef _compute_ecg(self, t_s: float) -> np.ndarray:

```

Returns:

np.ndarray of shape (12,) with voltages in mV

"""

```

dipole_3d = np.zeros(3) # X, Y, Z components

```

Retention window: only beats within last 0.85 s contribute

```

retention_window = 0.85

```

```

t_cutoff = t_s - retention_window

```

```
for beat in self._active_beats:
    if beat.sa_fire_time < t_cutoff:
        # Remove old beats to prevent memory growth
        self._active_beats.remove(beat)
        continue

    # Sum contributions from all nodes
    for node_name, t_act in beat.activation_times.items():
        if node_name not in self._nodes:
            continue

        node = self._nodes[node_name]

        # Depolarization Gaussian
        tau_depol = node.depol_duration # typically 12.7 ms
        g_depol = np.exp(-(t_s - t_act)**2 / (2 * tau_depol**2))
        if t_s < t_act:
            g_depol = 0

        # Repolarization Gaussian
        t_repol = t_act + node.repol_delay # peak at ~280 ms post-act
        tau_repol = node.repol_duration
        g_repol = np.exp(-(t_s - t_repol)**2 / (2 * tau_repol**2))

        # Dipole contribution
        dipole_node = node.dipole_vector * (
            g_depol - node.repol_amplitude * g_repol
        )
        dipole_3d += dipole_node

    # Project onto 12 lead vectors
    ecg_12 = self._forward_model.project(dipole_3d)
    return ecg_12
```

2.2 ConductionCable1D: Numerical Details

File: cardiac_sim/core/tissue/cable_1d.py (500 lines)

2.2.1 Cell Configuration

```

black# Fixed anatomical structure
_NV = 10 # Ventricular cells: His(0) + LBB(1-3) + LV(4-9)

# Cell segmentation (maps to nodes)
_ATR_SEGS = {
    'SA': (0, 1),          # SA node (index 0)
    'RA_EARLY': (1, 2),    # RA early
    'RA_MID': (2, 3),
    'RA_LATE': (3, 4),
    'LA_EARLY': (4, 5),    # LA early
    'LA_MID': (5, 6),
    'LA_LATE': (6, 8),     # LA late (indices 6-7)
}

_VEN_SEGS = {
    'HIS': (0, 1),
    'LBB_PROX': (1, 2),
    'LBB_MID': (2, 3),
    'LBB_DIST': (3, 4),
    'LV_SEPT': (4, 5),     # LV septum (early)
    'LV_ANT': (5, 6),
    'LV_LAT': (6, 7),      # LV lateral (late, ~645 ms)
    'LV_INF': (7, 8),
    'LV_POST': (8, 9),     # LV posterior
    'RV': (9, 10),        # Actually computed separately; not in sequential cable
}

```

2.2.2 Diffusion Coefficients (Phase 3-D Calibrated)

```

black# Atrial cable
0.0,      # SA node (point source, no diffusion)
2.88,     # RA
2.88,
2.88,
2.88,     # LA
2.88,
2.88,
2.88

```



```
])
```

```
# Ventricular cable
```

```
_D_VEN = np.array([
    0.0,      # His bundle (point source)
    72.0,     # LBB proximal (fast conduction, D_LV=3 earlier made QRS too wide)
    72.0,     # LBB mid
    72.0,     # LBB distal
    3.0,      # LV begins
    3.0,      # LV anterior
    3.0,      # LV lateral
    3.0,      # LV inferior
    3.0,      # LV posterior
    3.0       # RV auxiliary
])
```

```
# CFL check at initialization:
```

```
D_max_atr = 2.88
```

```
D_max_ven = 72.0
```

```
dt_int = 0.0001    # 0.1 ms micro-step
```

```
tau = 0.015        # 15 ms physical time scale
```

```
CFL_atr = D_max_atr * dt_int / (tau * 1**2) = 2.88 * 0.0001 / 0.015 = 0.0192
```

```
CFL_ven = D_max_ven * dt_int / (tau * 1**2) = 72.0 * 0.0001 / 0.015 = 0.480    (margin)
```

2.2.3 FHN State Variables

```
blackclass ConductionCable1D:
```

```
    # Atrial state: (cells, [voltage, recovery])
```

```
    self._sa = np.zeros((_NA, 2)) # Shape: (8, 2)
```

```
    self._sv = np.zeros((_NV, 2)) # Shape: (10, 2)
```

```
    # Activation tracking
```

```
    self._t_act_a = np.full(_NA, math.inf) # activation time per cell
```

```
    self._t_act_v = np.full(_NV, math.inf)
```

```
    # Internal clock
```

```
    self._t = 0.0
```

```

# Stimulus timing
self._stim_end_a = 0.0
self._av_pending_at = math.inf # when AV should fire

# Pathology state
self._av_conductance = params.av_node.conductance # 0-1
self._sep_conductance = params.ventricular.septal_conductance
self._rbb_conductance = params.ventricular.rbb_conductance
self._lbb_conductance = params.ventricular.lbb_conductance

```

2.2.4 New Beat: `new_beat`

```

blackdef new_beat(self, sa_fire_time: float, params: SimulationParameters) -> None:

```

```

    Initial conditions: all cells at rest, with stimulus applied
    to SA node.

```

```

    """

```

```

    self._t = sa_fire_time

```

```

# Initialization

```

```

V_REST, W_REST = FitzHughNagumoCell.get_resting_state()

```

```

self._sa[:, 0] = V_REST # voltage

```

```

self._sa[:, 1] = W_REST # recovery

```

```

self._sv[:, 0] = V_REST

```

```

self._sv[:, 1] = W_REST

```

```

# Mark activation times

```

```

self._t_act_a[:] = math.inf

```

```

self._t_act_v[:] = math.inf

```

```

self._t_act_a[0] = sa_fire_time # SA node fires at designated time

```

```

# Stimulus window

```

```

self._stim_end_a = sa_fire_time + _STIM_DUR

```

```

self._av_pending_at = math.inf

```

```

# Refractory enforcement

```

```

self._last_activation_v = np.full(_NV, -math.inf)

```

2.2.5 Integration Step: advance

```
blackdef advance(self, dt_s: float) -> dict[str, float]:
```

Returns:

dict mapping node names to activation times for nodes that crossed threshold during this step.

```
"""
n_micro = max(1, round(dt_s / _DT_INT))
dt_micro = dt_s / n_micro
newly_activated = {}

for _ in range(n_micro):
    self._t += dt_micro

    # Atrial and ventricular integration
    self._step_atrial(dt_micro)
    self._step_ventricular(dt_micro)

    # Detect threshold crossings (activation)
    for node_name, (i0, i1) in _ATR_SEGS.items():
        v_max = np.max(self._sa[i0:i1, 0])

        # If crossed threshold and not yet activated
        if v_max > V_THRESHOLD and self._t_act_a[i0] == math.inf:
            self._t_act_a[i0] = self._t # Record activation time
            newly_activated[node_name] = self._t

    # Similar for ventricular nodes
    ...

return newly_activated
```

2.2.6 Numerical Integration Functions

```
blackdef _step_atrial(self, dt: float) -> None:
    V = self._sa[:, 0].copy()
    W = self._sa[:, 1].copy()

    # Stimulus current
```

```

I = np.zeros(_NA)
if self._t < self._stim_end_a:
    I[0] = _STIM_AMP # Stimulus only at SA node

# FHN step
V_new, W_new = _fhn_cable_step(
    V, W, _D_ATR, I, dt,
    _TAU, _A, _B, _EPS
)

self._sa[:, 0] = V_new
self._sa[:, 1] = W_new

@numba.njit(cache=True) # JIT compile if Numba available
def _fhn_cable_step(V, W, D, I_ext, dt, tau, a, b, eps):
    """FHN dynamics on cable.

    Implements discretized equations (1)-(2) with finite-difference Laplacian.
    """
    N = len(V)

    # Laplacian (no-flux boundary conditions)
    lap = np.zeros(N)
    lap[1:-1] = V[:-2] - 2.0 * V[1:-1] + V[2:]
    lap[0]     = V[1] - V[0]
    lap[-1]    = V[-2] - V[-1]

    # FHN dynamics
    inv_tau = 1.0 / tau
    dv_dt = (D * lap + V - V**3 / 3.0 - W + I_ext) * inv_tau
    dw_dt = eps * (V + a - b * W) * inv_tau

    return V + dt * dv_dt, W + dt * dw_dt

```

Note: Numba JIT attempt occurs at import time; if incompatible NumPy version, falls back to pure NumPy (3–5× slower but functional).

```
language=Python, escapechar= basicstyle=, breaklines=true, keywordstyle=blue,  
commentstyle=gray, stringstyle=red, showstringspaces=false, frame=single,  
rulecolor=black!30
```

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Chapter 1

Pathological ECG Patterns: Clinical Descriptions and Reproduction

This chapter describes every pathological ECG pattern producible by the simulator, with clinical meaning and step-by-step reproduction instructions for each.

1.1 AV Conduction Abnormalities

1.1.1 First-Degree AV Block

Clinical Description

First-degree AV block is a **conduction delay** through the AV node, resulting in prolonged interval (> 200 ms), but with maintained 1:1 atrial-to-ventricular conduction. All P waves are followed by QRS complexes; no beats are dropped.

ECG Characteristics

- **PR interval:** Prolonged (typically 200–300 ms; normal is 120–200 ms)
- **QRS duration:** Normal (< 120 ms)
- **QRS morphology:** Normal (unless concurrent bundle branch block)
- **Heart rhythm:** Regular
- **P wave:** Normal morphology, slightly earlier than normal

Simulator Parameters: Reproduction

1. **Engine:** Use Phase 1/2 (faster, more stable pathology models)
2. **Pathology Panel:** Check “First-Degree AV Block”

3. **Parameter Adjustment** (manual, if needed):

- Navigate to **Parameters** panel → **AV Node** section
- **AV Delay**: Increase from default 100 ms to 150–200 ms
- Observe PR interval lengthening in real-time

4. **Expected Output**:

- P wave at normal time
- PR interval visibly longer than normal (e.g., 200 ms vs. 160 ms)
- QRS narrow and normal
- Repeats every cycle

Implementation

File: cardiac_sim/plugins/av_blocks.py

```
blackclass FirstDegreeAVBlock(AbstractPathologyPlugin):
    def name(self) -> str:
        return "First-Degree AV Block"

    def apply(self, engine: SimulationEngine, params: SimulationParameters) -> None:
        self._original_av_delay = params.av_node.av_delay_ms
        params.av_node.av_delay_ms = 150.0 # Increase by 50 ms
        engine.apply_pathology(self)

    def unapply(self, engine: SimulationEngine, params: SimulationParameters) -> None:
        params.av_node.av_delay_ms = self._original_av_delay
        engine.apply_pathology(None)
```

1.1.2 **Second-Degree AV Block (Mobitz II)**

Clinical Description

Mobitz II AV block involves **occasional dropped beats**. Some atrial activations fail to conduct to the ventricles; these manifest as P waves without following QRS complexes.

Unlike Mobitz I (progressive PR prolongation), Mobitz II has:

- Constant interval when conduction occurs
- Suddenly dropped beat(s) with no prior warning
- Often indicates infra-nodal conduction disease (His-Purkinje system)
- More dangerous (risk of complete heart block)

ECG Characteristics

- **PR interval:** Constant (normal or mildly prolonged)
- **Dropped beats:** Periodic (e.g., 2:1, 3:1 conduction ratios)
- **Appearance:** P wave without QRS; irregular rhythm
- **QRS duration:** May be widened (if His-Purkinje disease)

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2
2. **Pathology Panel:** Check “Second-Degree AV Block (Mobitz II)”
3. **Alternative Manual Method:**
 - **Parameters** → **AV Node** section
 - Set **AV Conductance** = 0.5 (50% conduction; some beats blocked)
 - May need to adjust **AV Refractory Period** to 400 ms
4. **Expected Output:**
 - Regular P wave rhythm (atrial activity unchanged)
 - Every 2nd, 3rd, or variable P wave followed by QRS
 - Dropped beats appearing as P waves alone

1.1.3 Third-Degree AV Block (Complete Heart Block)

Clinical Description

Complete AV block is **total failure of AV conduction**. Atria and ventricles beat independently (“AV dissociation”):

- Atria: Normal sinus rhythm, 60–80 bpm
- Ventricles: Escape rhythm (junctional or idioventricular) at much slower rate, 30–40 bpm

ECG Characteristics

- **P waves:** Regular, independent of QRS; may appear anywhere in the ECG
- **QRS complexes:** Slow rate (30–60 bpm), wide morphology
- **PR relationship:** Completely variable (sometimes short, sometimes P in QRS, sometimes after QRS)
- **Axis:** May show left axis deviation or extreme axis (if escape rhythm from LV)

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 (recommended; cleaner demonstration)
2. **Pathology Panel:** Check “Third-Degree AV Block (Complete)”
3. **Or Manual Method:**
 - **Parameters** → **AV Node**
 - Set **AV Conductance** = 0.0 (complete block)
 - Set **Ventricular** → **Ectopic Trigger Rate** to 40 bpm (escape rhythm)
4. **Expected Output:**
 - Atrial rhythm regular at normal rate (70 bpm)
 - Ventricular rhythm regular at escape rate (40 bpm)
 - P waves and QRS complexes dissociated
 - QRS wide (>120 ms) if escape from ventricular tissue

1.2 Bundle Branch Blocks

1.2.1 Right Bundle Branch Block (RBBB)

Clinical Description

Conduction block in the **right bundle branch** (RBB) forces RV activation to occur via **collateral conduction through the left ventricle**. This delays RV activation, causing:

- Widened (>120 ms)
- Characteristic **terminal R’ wave** (second positive deflection) in lead V1
- S wave in leads I and V5-V6

ECG Characteristics

- **QRS duration:** ≥ 120 ms (often 120–150 ms)
- **Lead V1:** Characteristic “M” or “rSR’” pattern (small r, deep S, tall R’)
- **Leads I, V5-V6:** S wave (negative deflection)
- **PR interval:** Normal
- **Axis:** Normal or rightward

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 or Phase 3
2. **Pathology Panel:** Check “Right Bundle Branch Block”
3. **Manual Method:**
 - **Parameters** $\rightarrow$ **Ventricular**
 - Set **RBB Conductance** = 0.0
 - Observe V1 lead: R’ should become visible
4. **Expected Timing** (Phase 3):
 - RV activation delayed to ≈ 645 ms (vs. ≈ 435 ms normal)
 - QRS broadened to ≈ 120 ms (vs. ≈ 70 ms)
 - T wave morphology also affected (opposite direction)

1.2.2 Left Bundle Branch Block (LBBB)

Clinical Description

Block in the **left bundle branch** delays LV activation, forcing it to propagate from RV via collaterals. This causes:

- Widened (>120 ms, often 140–160 ms)
- Broad, notched positive deflection in lateral leads (I, aVL, V5-V6)
- Absent Q wave in lateral/inferior leads (activation from RV)
- Characteristic “W” pattern in lead V1

ECG Characteristics

- **QRS duration:** ≥ 120 ms (typically 140–160 ms)
- **Leads I, V5-V6:** Broad positive (R) wave with notch or split
- **Lead V1:** W pattern (rS or QS)
- **Absent Q waves:** In leads I, aVL, V5-V6 (no normal Q wave in these leads)
- **Axis:** Often leftward (left axis deviation)

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 or Phase 3
2. **Pathology Panel:** Check “Left Bundle Branch Block”
3. **Manual Method:**
 - **Parameters** $\rightarrow$ **Ventricular**
 - Set **LBB Conductance** = 0.0
 - Observe lateral leads: broadened R wave with notch
4. **Expected Timing** (Phase 3):
 - LV activation delayed to ≈ 645 ms
 - QRS very broad (≈ 150 ms)
 - LV dominates morphology (rightward dipole from delayed LV)

1.3 Atrial Arrhythmias

1.3.1 Atrial Fibrillation (AFib)

Clinical Description

Atrial fibrillation is chaotic, uncoordinated electrical activity in the atria. The atrial rate exceeds 300 bpm with no organized conduction. Only occasional atrial activations penetrate the AV node, producing:

- Absent P waves (replaced by erratic baseline “fibrillatory waves”)
- Irregularly irregular ventricular rhythm
- Narrow complexes (if conduction intact below AV node)

ECG Characteristics

- **P waves:** Absent; baseline replaced by fine or coarse undulations (f waves)
- **Baseline:** Erratic, no isoelectric line
- **Ventricular rate:** Irregularly irregular (e.g., 60–100 bpm in controlled AFib)
- **QRS:** Narrow (<120 ms) unless concurrent BBB
- **RR intervals:** Variable (no two RR intervals the same)

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 (Phase 3 would require re-entrant circuit modeling)
2. **Pathology Panel:** Check “Atrial Fibrillation”
3. Implementation detail:
 - Atrial activation is replaced by chaotic re-entrant patterns
 - AV node conduction becomes probabilistic (some atrial waves blocked)
 - Baseline shows characteristic fibrillatory waves
4. **Expected Output:**
 - No P waves visible
 - Baseline wiggles (fine or coarse f waves)
 - QRS complexes at irregular intervals

1.3.2 Sinus Bradycardia

Clinical Description

Sinus bradycardia is a normal sinus rhythm at a rate < 60 bpm. All intervals and morphologies are normal; only the rate is slow.

Causes include athletic conditioning, hypothyroidism, increased vagal tone, or certain medications.

ECG Characteristics

- **Heart rate:** < 60 bpm (e.g., 50 bpm)
- **PR interval:** Normal
- **QRS:** Normal

- **Rhythm:** Regular
- **P-QRS-T morphology:** Completely normal; unchanged from NSR

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 or Phase 3
2. **Pathology Panel:** Check “Sinus Bradycardia”
3. **Manual Method:**
 - **Parameters** → **SA Node**
 - Set **Cycle Length (ms)** = 1200 (corresponds to 50 bpm)
 - Formula: $\text{HR (bpm)} = 60,000 / \text{Cycle Length (ms)}$
 - So 1200 ms → 50 bpm
4. **Expected Output:**
 - All complexes identical to normal
 - HR display shows 50 bpm
 - Horizontal spacing between complexes larger than normal

1.3.3 Sinus Tachycardia

Clinical Description

Sinus tachycardia is normal sinus rhythm at a rate > 100 bpm. All intervals and morphologies remain normal; only the rate is fast.

Causes include exercise, fever, anemia, hyperthyroidism, or sympathetic stimulation.

ECG Characteristics

- **Heart rate:** > 100 bpm (e.g., 140 bpm)
- **PR interval:** Normal
- **QRS:** Normal
- **Rhythm:** Regular
- **P wave:** May be peaked (increased amplitude) due to faster rate
- **T wave:** May slightly overlap preceding P wave

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 or Phase 3
2. **Pathology Panel:** Check “Sinus Tachycardia”
3. **Manual Method:**
 - **Parameters** → **SA Node**
 - Set **Cycle Length (ms)** = 400 (→ 150 bpm)
 - Formula: $60,000 / 400 = 150$ bpm
4. **Expected Output:**
 - All complexes identical to normal
 - HR display shows 150 bpm
 - Horizontal spacing between complexes narrower than normal
 - T wave may slightly overlap next P wave

1.4 Ventricular Arrhythmias

1.4.1 Premature Ventricular Contraction (PVC)

Clinical Description

A **PVC** is an ectopic ventricular beat that occurs earlier than expected in the cardiac cycle. On the ECG:

- Wide (>120 ms) with abnormal morphology
- Abnormal T wave (opposite polarity to QRS)
- Compensatory pause (next sinus beat occurs at normal time)
- Variable coupling interval (time from previous QRS to PVC)

ECG Characteristics

- **QRS duration:** >120 ms, often 140–180 ms
- **QRS morphology:** Abnormal; often broad and slurred
- **T wave:** Opposite polarity to QRS (discordant ST-T)
- **PR/PP interval:** No P wave before the PVC
- **Coupling interval:** Relatively fixed for multiple PVCs of same origin

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 (preferred) or Phase 3
2. **Pathology Panel:** Check “Premature Ventricular Contraction”
3. **Manual Method:**
 - **Parameters** → **Ventricular**
 - Set **Ectopic Trigger Rate (bpm)** = 60
 - Set **Ectopic Coupling Interval (ms)** = 600 (PVC fires 600 ms after normal beat)
4. **Expected Output:**
 - Normal sinus beats
 - Every so often, wide QRS with abnormal morphology
 - Compensatory pause after PVC (total RR interval = $2 \times$ sinus interval)

1.4.2 Ventricular Fibrillation (VF)

Clinical Description

Ventricular fibrillation is chaotic, uncoordinated electrical activity in the ventricles. There is no organized contractile function; the ECG shows:

- Erratic, low-amplitude oscillations (“fine VF”) or larger waves (“coarse VF”)
- No identifiable QRS, T, or ST segment
- Hemodynamic collapse (loss of perfusion)
- Lethal rhythm unless rapidly defibrillated

ECG Characteristics

- **Baseline:** Irregular undulations, no isoelectric segments
- **Amplitude:** Fine VF shows low amplitude (<1 mm); coarse VF shows >5 mm waves
- **Frequency:** 150–300 Hz baseline oscillations
- **QRS/T:** Absent
- **Rate:** Unmeasurable

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 (with special re-entrant modeling) or Phase 3
2. **Pathology Panel:** Check “Ventricular Fibrillation”
3. Implementation detail:
 - Ventricular conduction is randomized
 - Multiple competing depolarization foci activate ventricles chaotically
4. **Expected Output:**
 - Erratic baseline; no organized complexes
 - Fine or coarse undulations depending on rate of ectopic foci

Chapter 2

Architecture and Design Diagrams

2.1 Class Hierarchy Diagram

[Diagram - text description in document]

Figure 2.1: Class hierarchy and composition relationships. Abstract `SimulationEngine` is implemented by two concrete engines. `ConductionCableEngine` owns a `ConductionCable1D` and uses `LeadFieldModel`. GUI components interface with a worker thread.

2.2 Function Call Graph (Phase 3)

[Diagram - text description in document]

Figure 2.2: Main simulation loop call graph for Phase 3 engine. Each frame (~ 2 ms), `step` is called, which advances the cable via 20 micro-steps and computes the ECG dipole.

2.3 Data Flow: Parameter Changes

[Diagram - text description in document]

Figure 2.3: Data flow when user adjusts a parameter. The change propagates through the parameter model to the engine, which reinitializes and begins generating new ECG samples reflected in the display.

2.4 State Machine: Beat Lifecycle (Phase 3)

[Diagram - text description in document]

Figure 2.4: Beat state machine for `ConductionCableEngine`. A beat transitions from “No Beat” to “In Progress” when the SA node fires, then to “Committed” when all ventricular nodes have activated or a timeout is reached.

Chapter 3

Technical Reference and Appendices

3.1 FHN Model Constants and Ranges

Parameter	Phase 3-D Value	Notes
τ (time scale)	15 ms	Calibrated for correct P/PR/QRS timing
ε (recovery rate)	0.08	Controls APD; smaller = longer APD
a (FHN param)	0.7	Typical literature value
b (FHN param)	0.8	Typical literature value
Δt_{int} (micro-step)	0.1 ms	10 steps per output frame
N_{atr} (atrial cells)	8	SA + RA + LA
N_{ven} (ventricular cells)	10	His + LBB + LV
$D_{\text{atr,max}}$	2.88	Atrial diffusion
$D_{\text{ven,max}}$	72.0	Ventricular diffusion (His-LBB pathway)
D_{lv}	3.0	LV myocardial diffusion
V_{rest}	-1.2	Normalized rest potential
V_{peak}	+2.0	Normalized depolarization peak
$V_{\text{threshold}}$	-0.5	Activation threshold (normalized)

Table 3.1: FHN model constants for Phase 3-D (final calibration).

3.2 ECG Node Configuration (Phase 3)

Node	Depol σ (ms)	Repol Delay (ms)	Repol σ (ms)	Role
SA	12.7	280	50	SA node
RA	12.7	280	50	Right atrium
LA	12.7	280	50	Left atrium
AV	12.7	280	50	AV node
HIS	12.7	280	50	His bundle
LBB	12.7	280	50	L. bundle branch
RBB	12.7	280	50	R. bundle branch
SEPT	12.7	280	50	Septum
LV	40.0	280	50	L. ventricle (widened depol)
RV	12.7	280	50	R. ventricle

Table 3.2: ECG node Gaussian parameters. LV depol_duration widened to 40 ms so adjacent nodes produce overlapping Gaussians $\rightarrow$ smooth QRS morphology.

3.3 Installation and Running

3.3.1 Prerequisites

- Python 3.12+
- PyQt6 6.0
- NumPy 2.4 (Numba requires NumPy $\leq$ 2.4; we use 2.5 which forces fallback)
- pyqtgraph 0.13.0
- (Optional) Numba 0.59 for JIT compilation (requires compatible NumPy)

3.3.2 Setup Steps

```

black# Clone repository
cd ECG_simulator

# Create virtual environment
python3 -m venv .venv
source .venv/bin/activate # or .venv\textbackslashScripts\textbackslashactivate

# Install dependencies
pip install -r requirements.txt

```



```
# Run application
python main.py
```

3.3.3 Running Tests

```
black# Run all unit tests

# Run specific test file
python -m pytest tests/test_cable_model.py -v

# Run with coverage report
python -m pytest tests/ --cov=cardiac_sim --cov-report=html
```

pyth

3.4 Performance Considerations

3.4.1 Phase 1/2 (Graph Engine)

- Execution time per step: ≈ 0.1 ms
- Bottleneck: BFS graph traversal and Gaussian dipole summation
- Scales well to 1000+ Hz

3.4.2 Phase 3 (Cable Engine)

- Execution time per step: $\approx 1\text{--}2$ ms (on modern CPU)
- Simulated time per wall-clock second: $\approx 2.2\times$ real-time
- Bottleneck: $N \times M$ FHN integration (cables + micro-steps)
- Numba JIT (if available): $\approx 10\text{--}20\times$ speedup possible
- Current: NumPy fallback due to NumPy 2.5 incompatibility

3.5 Validation Results

3.5.1 Phase 3-D Timing Validation

Interval	Simulator	Clinical Target
P wave duration	64 ms	40–100 ms
PR interval	196 ms	120–200 ms
QRS duration	70 ms	<120 ms
QT interval	360 ms	300–440 ms
Heart rate	70 bpm	60–100 bpm

Table 3.3: Phase 3-D validated timing intervals compared to clinical targets (normal sinus rhythm at 70 bpm).

3.5.2 Multi-Beat Rhythm (Phase 3)

Successfully generates 6 consecutive QRS complexes at 70 bpm over 5 seconds without numerical artifacts or premature termination.

3.5.3 Amplitude Validation (Phase 3)

- QRS peak-to-trough: ≈ 1.3 mV (clinical: 0.5–2.5 mV)
- T wave amplitude: ≈ 0.6 mV
- QRS/T ratio: ≈ 2.2 (clinical normal: 1–2.5)

3.6 File Organization Summary

File Path	Purpose	Lines
main.py	Application entry point	50
cardiac_sim/core/parameter_model.py	Parameter dataclasses	200
cardiac_sim/core/interfaces.py	Abstract base classes	80
cardiac_sim/core/tissue/cable_1d.py	FHN cable implementation	500
cardiac_sim/core/conduction/graph.py	Graph structure	150
cardiac_sim/core/cell_models/fitzhugh_nagumo.py	FHN cell model	100
cardiac_sim/core/ecg/lead_field.py	Lead vectors	120
cardiac_sim/core/simulation_worker.py	Threading wrapper	150
cardiac_sim/simulation/graph_engine.py	Phase 1/2 engine	400
cardiac_sim/simulation/cable_engine.py	Phase 3 engine	450

File Path	Purpose	Lines
cardiac_sim/gui/main_window.py	Main GUI window	250
cardiac_sim/gui/parameter_panel.py	Parameter tree UI	180
cardiac_sim/gui/pathology_panel.py	Pathology selector	150
cardiac_sim/gui/ecg_display.py	ECG plotter	200
cardiac_sim/plugins/av_blocks.py	AV block pathologies	100
cardiac_sim/plugins/bundle_blocks.py	BBB pathologies	80
cardiac_sim/plugins/atrial.py	Atrial pathologies	100
cardiac_sim/plugins/ventricular.py	Ventricular pathologies	120
cardiac_sim/plugins/sinus_rhythms.py	Sinus rhythm variants	80
Total		≈ 3500

3.7 Glossary of Abbreviations

Abbreviation	Meaning
SA	Sinoatrial (atrial pacemaker)
AV	Atrioventricular (AV node)
RA/LA	Right/Left atrium
RV/LV	Right/Left ventricle
RBBB/LBBB	Right/Left bundle branch block
PR	Atrial to ventricular conduction time
QRS	Ventricular depolarization duration
ST	Repolarization segment
ECG	Electrocardiogram
APD	Action potential duration
CV	Conduction velocity
FHN	FitzHugh-Nagumo (ionic model)
PVC	Premature ventricular contraction
AFib	Atrial fibrillation
VF	Ventricular fibrillation

3.8 References and Further Reading

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5. Maringhini, G., et al. (2015). ECG simulation using cellular automata. *Comput. Biol. Med.*, 65, 207–215.

ECG Simulator Documentation
Part 4: GUI Reference & Parameter Reference

Comprehensive Technical Documentation

June 2026

Contents

1 Main Window Architecture

1.1 Overview

The PyQt6-based GUI provides real-time visualization of 12-lead ECG with interactive parameter control. The main window is organized into three primary regions:

- **Control Panel** (left): Heart rate, engine selection, beat controls
- **ECG Display** (center): Real-time scrolling 12-lead ECG trace
- **Parameter Panel** (right): 50+ physiological parameter controls

1.2 Control Panel Layout

1.2.1 Heart Rate (HR) Control

- **Widget Type:** QSpinBox
- **Range:** 30–220 bpm
- **Default:** 70 bpm
- **Step:** 1 bpm
- **Effect:** Directly sets SA node cycle length via:

$$\text{cycle_length} = \frac{60}{HR} \quad (1)$$

- **Clinical Range:** 60–100 bpm (normal sinus rhythm)
- **Bradycardia:** $\leq$ 60 bpm (typically 40–60 in athletes)
- **Tachycardia:** $\geq$ 100 bpm (up to 200+ during exercise)

1.2.2 Simulation Engine Selector

- **Widget Type:** QComboBox
- **Options:**
 - **graph** (Phase 1/2: BFS-based, $\leq$ 1000 Hz, deterministic)
 - **cable** (Phase 3: FHN cable, 2.2× real-time, physiologically accurate)
- **Default:** graph
- **Effect:** Switches underlying simulation engine
- **Behavior:** Resets current beat on switch

1.2.3 Beat Control Buttons

- **Start Button:** Begin continuous rhythm generation
 - Starts simulation timer (33 ms per frame @ 30 FPS)
 - Disables beat controls during active simulation
- **Stop Button:** Pause simulation (preserves current ECG trace)
 - Allows parameter adjustment without clearing ECG
- **Reset Button:** Clear ECG trace and restart
 - Clears all accumulated beats from display
 - Resets internal beat counters

2 ECG Display Widget

2.1 Real-Time 12-Lead Display

- **Library:** pyqtgraph (for real-time plotting performance)
- **Update Rate:** 30 FPS
- **Scroll Behavior:** Left-to-right, continuous scroll
- **Time Window:** 10 seconds visible (adjustable)
- **Leads Displayed:** All 12 ECG leads (6 frontal + 6 precordial)

2.2 12-Lead Organization

2.2.1 Frontal Leads (rows 1–2)

Lead	Axis	Clinical Use
I	0° (left)	Left-right forces
II	60° (inferolateral)	Inferior/lateral forces
III	120° (inferior)	Inferior forces
aVR	-150° (right)	Right atrium/ventricle
aVL	-30° (left)	Left ventricle
aVF	90° (inferior)	Inferior wall

2.2.2 Precordial Leads (rows 3–4)

Lead	Anatomical Position	Chamber
V1	4th ICS, right sternal border	RV outflow
V2	4th ICS, left sternal border	LV outflow
V3	Midway V2–V4	LV septum
V4	5th ICS, midclavicular	LV apex
V5	5th ICS, anterior axillary	LV lateral
V6	5th ICS, midaxillary	LV lateral

3 Parameter Panel Organization

3.1 Structure

The parameter panel is organized into 12 collapsible groups:

1. SA Node Parameters
2. AV Node Parameters
3. His Bundle & Purkinje
4. Atrial Conduction
5. Ventricular Conduction
6. Ventricular Refractory Periods
7. Escape Pacemaker
8. Atrial Fibrillation
9. Ectopic Focus
10. ECG Scaling
11. Plugin Controls
12. Advanced/Debug

4 Complete Parameter Reference (50+ Parameters)

4.1 Group 1: SA Node Parameters

Parameter	Description	Range	Unit
cycle_length	Interval between SA fires	0.3–2.0	s
funny_current	HCN4 channel conductance	0.0–1.0	g/g_max
autonomic_tone	Sympathetic/parasympathetic balance	-1.0 to +1.0	normalized

4.1.1 Usage Notes

- **cycle_length**: Typically set via HR spinner; direct control available
- **funny_current**: Alters SA pacemaker rate
 - 0.0 = No spontaneous depolarization (requires external stimulus)
 - 1.0 = Maximum rate (200 bpm)
- **autonomic_tone**: Modulates HR via sympathetic/parasympathetic effects
 - -1.0 = Maximum parasympathetic (vagal): 40 bpm
 - 0.0 = Balanced (normal): 70 bpm
 - +1.0 = Maximum sympathetic: 120 bpm

4.2 Group 2: AV Node Parameters

Parameter	Description	Range	Unit
av_delay	Conduction time through AV node	0.05–0.30	s
av_refractory	AV node absolute refractory period	0.20–0.40	s
conductance	AV nodal conductance (0=block)	0.0–1.0	g/g_max

4.2.1 Clinical Significance

- **av_delay**: Normal = 100–200 ms
 - Controls PR interval via: $PR = P\text{-wave} + av_delay + QRS$
 - Prolonged PR (First-degree AV block) when ≥ 200 ms
- **conductance**: Controls AV block severity
 - 1.0 = Normal conduction (1:1 conduction ratio)
 - 0.5 = 2:1 AV block (every 2nd atrial beat blocked)
 - 0.0 = Complete AV block (III° AV block)

4.3 Group 3: His Bundle & Purkinje System

Parameter	Description	Range	Unit
his_delay	His bundle conduction time	0.01–0.05	s
lbb_conductance	Left bundle branch conductance	0.0–1.0	g/g_max
rbb_conductance	Right bundle branch conductance	0.0–1.0	g/g_max
lahb_conductance	Left ant. fascicle conductance	0.0–1.0	g/g_max
lphb_conductance	Left post. fascicle conductance	0.0–1.0	g/g_max

4.3.1 Bundle Branch Block Patterns

- **LBBB** (Left Bundle Branch Block):
 - Set `lbb_conductance` = 0.0, keep `rbb_conductance` = 1.0
 - ECG: Broad QRS (≥ 120 ms), M-shaped in V5–V6
- **RBBB** (Right Bundle Branch Block):
 - Set `rbb_conductance` = 0.0, keep `lbb_conductance` = 1.0
 - ECG: Broad QRS (≥ 120 ms), M-shaped in V1–V2
- **LAHB** (Left Anterior Hemiblock):
 - Set `lahb_conductance` = 0.0, keep `lphb_conductance` = 1.0
 - ECG: Left axis deviation, QRS ≥ 120 ms
- **LPHB** (Left Posterior Hemiblock):
 - Set `lphb_conductance` = 0.0, keep `lahb_conductance` = 1.0
 - ECG: Right axis deviation, QRS ≥ 120 ms

4.4 Group 4: Atrial Conduction

Parameter	Description	Range	Unit
<code>ra_conduction</code>	Right atrial conduction velocity	0.50–1.50	m/s
<code>la_conduction</code>	Left atrial conduction velocity	0.50–1.50	m/s
<code>atrial_refractory</code>	Atrial myocardium absolute refractory	0.20–0.35	s

4.5 Group 5: Ventricular Conduction

Parameter	Description	Range	Unit
<code>sept_conduction</code>	Septal conduction velocity	0.3–1.5	m/s
<code>lv_conduction</code>	LV free wall conduction velocity	0.30–1.0	m/s
<code>rv_conduction</code>	RV free wall conduction velocity	0.30–1.0	m/s

4.5.1 QRS Duration Formula

Total QRS duration depends on conduction velocities and geometric path lengths. For Phase 1/2 engine:

$$QRS_{\text{duration}} \approx \frac{\text{longest path}}{v_{\min}} \quad (2)$$

4.6 Group 6: Ventricular Refractory Periods

Parameter	Description	Range	Unit
<code>vent_abs_refr</code>	Absolute refractory period	0.25–0.45	s
<code>vent_rel_refr</code>	Relative refractory period	0.10–0.35	s

4.6.1 Physiological Basis

- **Absolute Refractory Period (ARP):** Voltage-gated Na^+ channels inactivated
 - No stimulus can trigger action potential
 - Normal duration: 200–300 ms
 - Set on Phase 1/2 engine only
- **Relative Refractory Period (RRP):** Partial recovery of Na^+ channel availability
 - Stimulus can trigger AP but with reduced amplitude
 - Conduction may be slower or blocked
 - Normal duration: 150–300 ms additional
- **Phase 3 Engine:** Refractory periods emerge naturally from FHN w variable
 - Recovery time determined by:

$$\tau_w = \frac{\tau_s}{b \cdot \varepsilon} = \frac{0.015}{0.8 \times 0.08} \approx 234 \text{ ms} \quad (3)$$

- No explicit refractory parameter needed

4.7 Group 7: Escape Pacemaker

Parameter	Description	Range	Unit
escape_enabled	Enable backup pacemaker	Yes/No	bool
escape_interval	Escape interval (interval without SA)	0.5–3.0	s
escape_origin	Ectopic site (AV/HIS/LV)	enum	-

4.7.1 Escape Pacemaker Mechanism

- Automatically fires if no atrial impulse received
- Clinical significance:
 - **Junctional escape** (origin=AV): Fires at 40–60 bpm, narrow QRS
 - **Ventricular escape** (origin=LV): Fires at 20–40 bpm, broad QRS
- escape_interval determines delay before firing:
 - If no stimulus received for $\geq$ escape_interval seconds, escape fires
 - Common clinical values: 1.5–3.0 seconds

4.8 Group 8: Atrial Fibrillation

Parameter	Description	Range	Unit
af_enabled	Enable atrial fibrillation	Yes/No	bool
af_rate	F-wave (fibrillatory wave) rate	300–700	bpm
af_rr_distribution	RR interval variation type	fixed/poisson/gaussian	enum
af_f_amplitude	F-wave amplitude	0.01–0.20	mV

4.8.1 AF ECG Characteristics

- **AF Rate:** Normal 300–600 bpm (fibrillatory waves from disorganized reentry)
 - Slower rates (300 bpm) suggest possible organized activity (flutter)
 - Faster rates (600+ bpm) suggest rapid fibrillatory activity
- **RR Distribution:**
 - fixed: Constant interval (used for testing)
 - poisson: Random exponential intervals (physiologic, irregular)
 - gaussian: Normal distribution around mean (moderate irregularity)
- **F-wave Amplitude:** Visible baseline oscillations
 - Coarse AF: ≥ 0.10 mV (more organized)
 - Fine AF: ≤ 0.05 mV (more disorganized)

4.9 Group 9: Ectopic Focus

Parameter	Description	Range	Unit
ectopic_enabled	Enable ectopic focus	Yes/No	bool
ectopic_coupling	Coupling interval (PVC timing)	0.40–0.80	s
ectopic_repetitive	Consecutive ectopics	Yes/No	bool
ectopic_origin	Ectopic site	LA/RA/LV/RV	enum

4.9.1 Premature Contraction Types

- **PAC (Premature Atrial Contraction):**
 - Set ectopic_origin = LA or RA
 - Normal coupling: 350–450 ms
 - Early coupling (≤ 350 ms) may block AV node $\rightarrow$ aberrant conduction
- **PVC (Premature Ventricular Contraction):**
 - Set ectopic_origin = LV or RV
 - Normal coupling: 400–600 ms
 - ECG: Broad QRS (≥ 120 ms), T-wave opposite normal polarity
 - Couplets/triplets possible if ectopic_repetitive = true

4.10 Group 10: ECG Scaling

Parameter	Description	Range	Unit
amplitude_scale	Global amplitude multiplier	0.1–5.0	unitless
baseline_shift	DC offset for visibility	-2.0–+2.0	mV

5 Advanced Controls & Debug Parameters

5.1 Group 11: Plugin Controls

- **Active Plugins:** Dropdown showing enabled pathologies
- **Plugin Stacking:** Select multiple pathologies simultaneously
- **Save/Load Presets:** Store/retrieve parameter combinations

5.2 Group 12: Advanced/Debug

Parameter	Description	Range	Unit
debug_mode	Enable diagnostic output	Yes/No	bool
show_activation_times	Print node activation times	Yes/No	bool
cable_internal_dt	Phase 3 integration step	0.0001–0.001	s

6 Parameter Adjustment Workflow

6.1 Typical Use Cases

6.1.1 Scenario 1: Simulate Normal Sinus Rhythm

1. HR = 70 bpm
2. All conductances = 1.0
3. All refractories = defaults
4. af_enabled = false
5. ectopic_enabled = false

6.1.2 Scenario 2: Simulate First-Degree AV Block

1. HR = 70 bpm
2. av_delay = 0.25 s (prolonged from normal 0.15 s)
3. av_conductance = 1.0 (1:1 conduction maintained)
4. Result: PR interval ≥ 200 ms

6.1.3 Scenario 3: Simulate Left Bundle Branch Block

1. HR = 70 bpm
2. lbb_conductance = 0.0 (complete block)
3. rbb_conductance = 1.0 (normal)
4. Result: Broad QRS (≥ 120 ms), wide S wave in V5–V6

6.1.4 Scenario 4: Simulate Atrial Fibrillation

1. HR = 130 bpm (controlled ventricular rate)
2. af_enabled = true
3. af_rate = 450 bpm (fibrillatory waves)
4. af_rr_distribution = poisson
5. Result: Irregular rhythm, absent P waves, f-waves visible

7 Real-Time Display Updates

7.1 Refresh Mechanism

- GUI updates at 30 FPS
- ECG trace accumulates every 33 ms
- Leads redrawn continuously
- Parameters can be adjusted without stopping simulation

7.2 Performance Notes

- Phase 1/2 engine: 1000 Hz update rate → GPU-limited by display
- Phase 3 engine: 150 Hz (2.2× real-time) → smooth real-time performance
- Parameter changes take effect immediately on next beat
- HR changes cause immediate cycle_length update

ECG Simulator Documentation

Part 5: Pathological ECG Patterns

Complete Catalog of Cardiac Arrhythmias & Conduction Abnormalities

June 2026

Contents

1 AV Block Syndrome (Group 1: Conduction Delay/Block)

1.1 First-Degree AV Block (I°)

1.1.1 Clinical Definition

Prolongation of AV conduction interval beyond upper limit of normal (200 ms), with 1:1 conduction maintained. No impulses are blocked; all atrial impulses conduct to ventricles, but with delay.

1.1.2 ECG Morphology

- **PR Interval:** ≥ 200 ms (abnormal, normal 120–200 ms)
- **P Waves:** Normal morphology and rate
- **QRS Complex:** Normal duration (≤ 120 ms)
- **Conduction Ratio:** 1:1 (every atrial beat conducts)
- **Rhythm:** Regular (no irregularity)

1.1.3 Physiological Cause

- Increased AV nodal delay (slower conduction through AV node)
- Increased refractoriness in AV nodal tissue
- Increased parasympathetic (vagal) tone
- Aging, fibrosis of AV node tissue

1.1.4 Clinical Significance

- Usually benign and asymptomatic
- May progress to higher-degree AV block in some patients
- Common in athletic individuals (vagal tone)
- Associated with: digitalis toxicity, beta-blockers, calcium channel blockers

1.1.5 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set av_delay = 0.25 s (normal 0.15 s, prolonged by 100 ms)
4. Set av_conductance = 1.0 (maintain 1:1 conduction)
5. Disable all other pathologies
6. Press Start
7. **Expected Result:** Visible PR interval ≥ 200 ms; narrow QRS; regular rhythm

1.2 Second-Degree AV Block, Mobitz Type I (Wenckebach)

1.2.1 ECG Morphology

- **PR Interval:** Progressively lengthens (200 ms $\rightarrow$ 300+ ms)
- **Dropped Beat Pattern:** P wave present but no QRS follows
- **Conduction Ratio:** Usually 5:4 or 4:3 (e.g., 5 P waves, 4 QRS complexes)
- **Rhythm:** Grouped beating; pause containing dropped beat
- **QRS Duration:** $\leq$ 120 ms (narrow QRS)

1.2.2 Physiological Cause

- Progressive fatigue of AV nodal conduction
- Decreasing AV nodal conductance over multiple cycles
- Each impulse arrives during relative refractory period
- Eventually one impulse fails to conduct (Wenckebach periodicity)

1.2.3 Clinical Significance

- Usually benign; rarely requires treatment
- Associated with: vagal tone, digitalis toxicity, ischemia
- Characteristically regular irregularity (predictable dropped beats)
- Progression to higher block is uncommon

1.2.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set av_conductance = 0.7 (partial reduction)
4. Set av_refractory = 0.35 s (slightly prolonged)
5. Gradually decrease av_conductance from 1.0 $\rightarrow$ 0.6 during observation
6. **Expected Result:** Progressive PR lengthening; periodic dropped beats; grouped rhythm

1.3 Second-Degree AV Block, Mobitz Type II

1.3.1 ECG Morphology

- **PR Interval:** Constant in conducted beats (no progressive lengthening)
- **Dropped Beats:** Sudden, without warning
- **Conduction Ratio:** Fixed (2:1, 3:1, or variable)
- **QRS Duration:** Usually $\leq$ 120 ms (wide QRS, indicating infranodal block)
- **Rhythm:** Regularly irregular (predictable dropped beats at fixed intervals)

1.3.2 Physiological Cause

- Block located in His bundle or bundle branches (infranodal)
- AV node conduction preserved; block occurs below AV node
- Abnormal automaticity or refractory period in conduction system
- Structural damage (ischemia, fibrosis, infiltration)

1.3.3 Clinical Significance

- **MORE SERIOUS than Mobitz I**
- Indicates structural disease of conduction system
- High risk of progression to complete AV block
- Often requires pacemaker implantation
- Associated with: acute MI, digitalis toxicity, cardiomyopathy

1.3.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set his_delay = 0.04 s (prolonged)
4. Set lbb_conductance = 0.5 (partial block simulates infranodal disease)
5. av_conductance = 1.0 (AV node normal; His bundle affected)
6. **Expected Result:** Fixed (not progressive) PR; sudden dropped beats; broad QRS

1.4 Third-Degree AV Block (Complete AV Block)

1.4.1 ECG Morphology

- **PR Interval:** Variable (atria and ventricles beat independently)
- **P Waves:** Normal rate and morphology (typically 60–100 bpm)
- **QRS Complexes:** Slow escape rhythm (40–60 if junctional, 20–40 if ventricular)
- **Rhythm:** Regular (atria regular; ventricles regular; unrelated to each other)
- **QRS Duration:** Depends on escape origin
 - Junctional escape: ≤ 120 ms (narrow)
 - Ventricular escape: ≥ 120 ms (broad)

1.4.2 Physiological Cause

- Complete dissociation between atrial and ventricular conduction
- Every atrial impulse blocked; no conduction through AV node or His bundle
- Ventricles must rely on backup (escape) pacemaker
- If escape fails $\rightarrow$ cardiac arrest

1.4.3 Clinical Significance

- **EMERGENT CONDITION**; requires immediate intervention
- Escape rate often inadequate for normal hemodynamics
- Symptoms: syncope, heart failure, shock
- Associated with: acute MI, digitalis toxicity, Lyme disease (tick-borne), infiltration
- Treatment: Temporary/permanent pacemaker

1.4.4 GUI Reproduction Steps

1. HR = 70 bpm (atrial rate)
2. Engine = graph (Phase 1/2)
3. Set av_conductance = 0.0 (complete block)
4. Set escape_enabled = true
5. Set escape_origin = HIS (junctional escape)
6. Set escape_interval = 1.5 s
7. **Expected Result:** P waves at 70 bpm; QRS at 40 bpm; complete dissociation

2 Bundle Branch Blocks (Group 2)

2.1 Left Bundle Branch Block (LBBB)

2.1.1 ECG Morphology

- **QRS Duration:** ≥ 120 ms (broad)
- **QRS Shape:** Notched/M-shaped in lateral leads (V5, V6, I, aVL)
- **ST Segment:** Sagging (downsloping from QRS)
- **T Waves:** Opposite to QRS polarity (typically negative in lateral leads)
- **Axis:** Usually normal to left

2.1.2 Physiological Cause

- Complete or near-complete block of left bundle branch
- Right bundle branch conducts normally
- RV depolarizes first (narrow septal Q wave)
- LV depolarizes slowly via collaterals $\rightarrow$ delayed and slow LV depolarization
- Resultant QRS vector points left and posteriorly

2.1.3 Clinical Significance

- Often indicates structural heart disease (MI, cardiomyopathy, fibrosis)
- May be associated with HF (reduced ejection fraction)
- Interferes with MI diagnosis (ST elevations may be masked)
- New LBBB = urgent evaluation required

2.1.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set lbb_conductance = 0.0 (complete block)
4. Set rbb_conductance = 1.0 (normal)
5. **Expected Result:** QRS ≥ 120 ms; M-shaped QRS in V5, V6; negative T waves in lateral leads

2.2 Right Bundle Branch Block (RBBB)

2.2.1 ECG Morphology

- **QRS Duration:** ≥ 120 ms (broad)
- **QRS Shape:** M-shaped ("rsR' pattern") in V1–V2 (septal, left posterior, left anterior)
- **S Wave:** Deep and wide in I, V5, V6
- **ST Segment:** May be elevated in V1–V2 (can mimic STEMI)
- **T Waves:** Usually negative in V1–V3

2.2.2 Physiological Cause

- Complete or near-complete block of right bundle branch
- Left bundle branch conducts normally
- LV depolarizes first
- RV depolarizes slowly via collaterals $\rightarrow$ delayed RV vector rightward
- Resultant QRS vector terminates rightward (terminal R' in V1)

2.2.3 Clinical Significance

- Often normal variant (RBBB seen in 0.1% of population)
- May indicate: PE, atrial septal defect, RV strain, anterior MI
- Less ominous than LBBB if isolated
- New RBBB with symptoms warrants evaluation

2.2.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set rbb_conductance = 0.0 (complete block)
4. Set lbb_conductance = 1.0 (normal)
5. **Expected Result:** QRS $\lesssim$ 120 ms; rsR' in V1–V2; deep S in I, V5–V6

2.3 Left Anterior Fascicular Block (LAHB)

2.3.1 ECG Morphology

- **Left Axis Deviation:** Extreme left axis (-45° to -90°)
- **QRS Duration:** Typically $\lesssim$ 120 ms (narrow, unlike bundle blocks)
- **QRS Pattern:** Small R in I, Q and S in III (indicating leftward vector)
- **Clinical Finding:** Lead I shows initial R, then S wave

2.3.2 Physiological Cause

- Block of left anterior fascicle (supplies anterosuperior LV wall)
- Right and left posterior fascicles conduct normally
- Right and posterior vectors cancel; anterosuperior vector dominates
- Resultant axis points strongly leftward and superiorly

2.3.3 Clinical Significance

- Often associated with: inferior MI, degenerative disease
- May progress to complete LBB block or higher-degree AV block
- Requires baseline ECG documentation; new LAHB may indicate acute disease

2.3.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set lahb_conductance = 0.0 (anterior fascicle blocked)
4. Set lphb_conductance = 1.0, rbb_conductance = 1.0 (other paths normal)
5. **Expected Result:** Extreme left axis deviation; narrow QRS; qS pattern in III

2.4 Left Posterior Fascicular Block (LPHB)

2.4.1 ECG Morphology

- **Right Axis Deviation:** Rightward axis ($+90^\circ$ to $+120^\circ$)
- **QRS Duration:** ≤ 120 ms (narrow)
- **QRS Pattern:** Small R in III, Q and S in I
- **Clinical Finding:** Rightmost QRS vector

2.4.2 Physiological Cause

- Block of left posterior fascicle (posteroinferior LV wall)
- Anterior and right fascicles conduct normally
- Inferior and posterior vectors cancel; anterior vector dominates
- Resultant axis points rightward

2.4.3 Clinical Significance

- Rare (much less common than LAHB)
- Associated with: inferior MI, infiltrative diseases
- Requires differentiation from RBBB (LPHB has narrow QRS)

2.4.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set lphb_conductance = 0.0 (posterior fascicle blocked)
4. Set lahb_conductance = 1.0, rbb_conductance = 1.0
5. **Expected Result:** Right axis deviation; narrow QRS; rS pattern in I

3 Sinus Rhythm Abnormalities (Group 3)

3.1 Sinus Bradycardia

3.1.1 ECG Morphology

- **Heart Rate:** ≤ 60 bpm (normal 60–100 bpm)
- **P Waves:** Normal morphology
- **PR Interval:** Normal (120–200 ms)
- **QRS Complex:** Normal (≤ 120 ms)
- **Rhythm:** Regular (sinus rhythm maintained)

3.1.2 Physiological Cause

- Decreased SA nodal automaticity
- Increased parasympathetic tone (vagal)
- Decreased sympathetic tone
- Decreased metabolic rate (sleep, hypothermia)

3.1.3 Clinical Significance

- Often normal in athletes (resting HR 40–50 bpm)
- May indicate: hypothyroidism, increased ICP, medication effects, ischemia
- Symptomatic bradycardia: dizziness, syncope, fatigue
- Severe (<40 bpm) may require pacemaker

3.1.4 GUI Reproduction Steps

1. Set HR = 45 bpm (via HR spinner)
2. Alternatively: Set autonomic_tone = -1.0 (maximum parasympathetic)
3. **Expected Result:** Regular rhythm at 45 bpm; normal PQRST morphology

3.2 Sinus Tachycardia

3.2.1 ECG Morphology

- **Heart Rate:** >100 bpm (normal 60–100 bpm)
- **P Waves:** Normal morphology (may be small if rate very high)
- **PR Interval:** Shortened as rate increases
- **QRS Complex:** Normal (<120 ms)
- **Rhythm:** Regular

3.2.2 Physiological Cause

- Increased SA nodal automaticity
- Increased sympathetic tone
- Increased metabolic demands
- Response to exercise, fever, pain, anxiety

3.2.3 Clinical Significance

- Often physiologic (exercise, emotion, fever)
- May indicate: hyperthyroidism, anemia, heart failure, sepsis, hypovolemia
- Persistent tachycardia at rest warrants investigation
- PR shortening is expected (not abnormal)

3.2.4 GUI Reproduction Steps

1. Set HR = 150 bpm (via HR spinner)
2. Alternatively: Set autonomic_tone = +1.0 (maximum sympathetic)
3. **Expected Result:** Regular rhythm at 150 bpm; normal morphology; shortened PR

4 Atrial Arrhythmias (Group 4)

4.1 Atrial Fibrillation (AF)

4.1.1 ECG Morphology

- **P Waves:** ABSENT (replaced by baseline undulation)
- **Baseline:** Irregular oscillation (fibrillatory waves, "f waves")
- **QRS Complex:** Normal morphology (≤ 120 ms), but occurs at irregular intervals
- **Rhythm:** Completely irregular ("irregularly irregular")
- **Ventricular Rate:** Typically 100–180 bpm (fast ventricular response)
- **f-Wave Amplitude:** Variable (coarse AF 1–5 mm; fine AF ≤ 1 mm)

4.1.2 Physiological Cause

- Rapid disorganized reentry throughout atria (300–600 bpm)
- Each atrial wavelet is refractory to others $\rightarrow$ chaotic activation
- AV node bombardment by multiple disorganized impulses
- AV node acts as gatekeeper $\rightarrow$ variable conduction to ventricles
- Result: Irregular ventricular response despite chaotic atrial rate

4.1.3 Clinical Significance

- **MOST COMMON CARDIAC ARRHYTHMIA**
- Associated with: heart failure, hypertension, valvular disease, hyperthyroidism, age
- Major risk: Stroke (from atrial thrombus), hemodynamic compromise
- Treatment: Rate control (beta-blockers, CCBs, digoxin), anticoagulation, rhythm control
- Symptom severity depends on: ventricular rate, HF status, exercise tolerance

4.1.4 GUI Reproduction Steps

1. HR = 130 bpm (target controlled ventricular rate)
2. Engine = graph (Phase 1/2) or cable (Phase 3)
3. Set af_enabled = true
4. Set af_rate = 450 bpm (fibrillatory wave rate)

5. Set `af_rr_distribution = poisson` (realistic irregular RR intervals)
6. Set `af_f_amplitude = 0.05–0.10 mV`
7. **Expected Result:** Absent P waves; baseline f-waves visible; irregular QRS intervals

4.2 Premature Atrial Contraction (PAC)

4.2.1 ECG Morphology

- **P Wave:** Early (abnormal morphology, axis different from sinus)
- **PR Interval:** May be prolonged if early impulse hits refractory AV node
- **QRS Complex:** Typically normal; may be aberrant if very early
- **Coupling Interval:** 300–600 ms (time from previous QRS to premature QRS)
- **Compensatory Pause:** May be full or incomplete

4.2.2 Physiological Cause

- Premature automaticity in atrial muscle (outside SA node)
- Enhanced normal automaticity or triggered activity
- Ectopic focus in LA or RA
- Reset of SA node rhythm (typically full compensatory pause)

4.2.3 Clinical Significance

- Often benign if isolated
- Frequency may herald atrial fibrillation or other arrhythmias
- Associated with: caffeine, stress, smoking, sleep deprivation, hypoxia
- Rare: Frequent PACs may indicate atrial disease or electrolyte abnormality

4.2.4 GUI Reproduction Steps

1. HR = 70 bpm (normal sinus rate)
2. Set `ectopic_enabled = true`
3. Set `ectopic_origin = LA or RA`
4. Set `ectopic_coupling = 0.50 s` (500 ms after normal QRS)
5. Set `ectopic_repetitive = false` (isolated PAC)
6. **Expected Result:** Early P wave with different morphology; QRS follows; pause to reset rhythm

5 Ventricular Arrhythmias (Group 5)

5.1 Premature Ventricular Contraction (PVC)

5.1.1 ECG Morphology

- **QRS Complex:** Broad (>120 ms), bizarre morphology
- **QRS Axis:** Abnormal (direction depends on ectopic origin)
- **T Wave:** Opposite to QRS polarity (discordant ST-T wave)
- **ST Segment:** Depressed or elevated (opposite to QRS)
- **Coupling Interval:** 300–600 ms (time from previous beat)
- **Compensatory Pause:** Usually full (reset SA node via retrograde conduction)

5.1.2 Physiological Cause

- Premature automaticity in ventricular muscle
- Ectopic focus in LV or RV
- Ventricular depolarization begins outside normal conduction system
- Slow spread via myocardium $\rightarrow$ broad, aberrant QRS

5.1.3 Clinical Significance

- Single or rare PVCs often benign
- Frequent PVCs ($>10/\text{min}$): May indicate structural disease or electrolyte abnormality
- Patterns:
 - **Bigeminy:** Every 2nd beat is PVC
 - **Trigeminy:** Every 3rd beat is PVC
 - **Couplet:** Two consecutive PVCs
 - **Triplet/NSVT:** 3 consecutive PVCs (nonsustained VT)
- Risk factors: Hypertension, CAD, cardiomyopathy, hypoxia, hypokalemia

5.1.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Set ectopic_enabled = true
3. Set ectopic_origin = LV or RV
4. Set ectopic_coupling = 0.50 s (500 ms)
5. Set ectopic_repetitive = false (single PVC)
6. **Expected Result:** Broad bizarre QRS; discordant T wave; full compensatory pause

5.2 Repetitive Ventricular Ectopy

5.2.1 Bigeminy Pattern

- **Pattern:** Normal beat — PVC — Normal beat — PVC — ...
- **Ratio:** 1:1 (equal number of normal and premature beats)
- **Rate:** If base is 70 bpm, bigeminy rate is 140 bpm
- **Clinical Significance:** Suggests more serious underlying process; warrants evaluation

5.2.2 Trigeminy Pattern

- **Pattern:** Normal — Normal — PVC — repeating
- **Ratio:** 2:1
- **Rate:** Lower than bigeminy

5.2.3 GUI Reproduction for Bigeminy

1. HR = 70 bpm
2. Set `ectopic_enabled = true`
3. Set `ectopic_origin = LV`
4. Set `ectopic_coupling = 0.50 s`
5. Set `ectopic_repetitive = true` (repetitive pattern)
6. **Expected Result:** Every 2nd beat is broad PVC; perfect 1:1 bigeminy

ECG Simulator Documentation

Part 6: System Architecture & Design

Module Organization, Class Hierarchy, Extension Points

June 2026

Contents

1 Three-Layer Architecture

1.1 Overview

The ECG simulator is organized into three distinct layers:

1. **Physics Layer:** Cardiac physiology (cell models, propagation, dipoles)
2. **Simulation Layer:** Algorithm selection (Phase 1/2 vs Phase 3), beat management
3. **GUI Layer:** PyQt6 interface, real-time visualization, parameter control

1.2 Layer Interaction

```

GUI Layer (main_window.py, parameter_panel.py)
    |
    | controls
    |
Simulation Layer (graph_engine.py, cable_engine.py)
    |
    | uses
    |
Physics Layer (cell_models/, tissue/, ecg/)

```

1.3 Data Flow

1. User adjusts HR or parameters in GUI
2. GUI updates ParameterModel (single source of truth)
3. SimulationEngine queries ParameterModel for each beat
4. Engine executes propagation algorithm
5. Forward model computes 12-lead ECG voltages
6. GUI displays voltages on pyqtgraph widget

2 Physics Layer (Core Modules)

2.1 Module Organization

Module	Purpose	Lines
cardiac_sim/core/cell_models/fitzhugh_nagumo.py	FHN cell model ODEs	150
cardiac_sim/core/tissue/cable_1d.py	1-D monodomain integration	500
cardiac_sim/core/tissue/graph.py	DAG + BFS conduction	350
cardiac_sim/core/ecg/lead_field.py	12-lead lead vectors	150

2.2 Class Hierarchy: Cell Models

```

BaseCellModel (abstract)
    FitzHughNagumoCell
    [Future: HodgkinHuxley, Courtemanche, tenTusscher]

```

Key Methods:

- get_initial_state()

- compute_derivatives(t, state)
- stimulate(t_start, duration, amplitude)
- v_to_mv(v_norm, v_rest, v_peak)

2.2.1 FitzHughNagumoCell Interface

```

blueclass FitzHughNagumoCell(BaseCellModel):
    bluedef get_initial_state(self) -> np.ndarray:
gray        gray#gray grayReturnsgray gray[grayV_RESTgray,gray grayW_RESTgray]
        bluereturn np.array([-1.200, -0.625])

    bluedef compute_derivatives(self, t, state):
gray        gray#gray graySolvesgray graydimensionlessgray grayFHNgray
        grayODEs
gray        gray#gray grayScalesgray graytogy grayphysicalgray graytimegray
        grayviagray grayTAU_S
        v, w = state
        dv = v - v**3/3 - w + I_ext
        dw = self.eps * (v + self.a - self.b * w)
        bluereturn np.array([dv, dw])

    bluedef v_to_mv(self, v_norm, v_rest, v_peak):
gray        gray#gray grayMapsgray gray[-1.2,gray gray2.0]gray gray    gray
        gray[-85,gray gray+30]gray graymV
        bluereturn v_rest + (v_peak - v_rest) * \
            (v_norm - self.V_REST) / (self.V_PEAK - self.V_REST)

```

2.3 Cable Model (1-D Tissue)

```

blueclass Cable1D:
    bluedef __init__(self, params):
gray        gray#gray grayInitializegray grayVgray,gray grayWgray graystate
        gray grayarraysgray gray(18gray graycellsgray)
        self.V = np.ones(18) * self.V_REST
        self.W = np.ones(18) * self.W_REST
        self._check_cfl()

    bluedef new_beat(self, sa_fire_time, params):
gray        gray#gray grayResetgray graystategray,gray graymarkgray graySAggray
        grayactivated
        self.V.fill(self.V_REST)
        self.W.fill(self.W_REST)
        self._active_beat_start = sa_fire_time

    bluedef advance(self, dt):
gray        gray#gray grayIntegrategray graydtgray graysecondsgray grayviagray
        grayn_microgray graysteps
gray        gray#gray grayReturnsgray graydictgray grayofgray graynewlygray
        grayactivatedgray graynodes
        bluefor _ bluein bluerange(n_micro):
            self._step_atrial(DT_INT)
            self._step_ventricular(DT_INT)
        bluereturn newly_activated_dict

```

3 Simulation Layer

3.1 SimulationEngine Interface

```
SimulationEngine (abstract)
    GraphEngine (Phase 1/2: BFS-based)
    CableEngine (Phase 3: FHN cable)
```

Key Methods:

```
- generate_beat(hr, params) -> dict[node -> t_act]
- get_activation_times() -> dict
- advance_time(dt)
```

3.2 GraphEngine (Phase 1/2)

BFS Algorithm:

1. Start queue with SA node (t=0)
2. For each node:
 - a. Compute conduction time to each neighbor
 - b. Check if neighbor is refractory
 - c. If NOT refractory: add to queue with new activation time
 - d. If refractory: skip
3. Return dict[node -> activation_time] for entire beat

Refractory Check:

```
t_candidate = t_current + conduction_time
if t_candidate < last_activation[node] + ERP[node]:
    # Skip (still refractory)
else:
    # Conduct
```

3.2.1 Key Implementation Details

```
bluedef run_bfs(self, sa_fire_time, params):
    activation_times = {}
    queue = deque([(SA_NODE, sa_fire_time)])

    bluewhile queue:
        current_node, t_curr = queue.popleft()
        activation_times[current_node] = t_curr

    gray    gray#gray grayFindgray grayrefractorygray graystatus
    last_act = self._last_activation.get(current_node, -inf)
    blueif t_curr < last_act + ERP[current_node]:
        bluecontinue gray gray#gray graySkipgray grayifgray graystillgray
            grayrefractory

    gray    gray#gray grayProcessgray grayneighbors
    bluefor neighbor, cond_time bluein self.adjacency[current_node]:
        t_next = t_curr + cond_time
        queue.append((neighbor, t_next))

    bluereturn activation_times
```

3.3 CableEngine (Phase 3)

Phase 3: FHN Cable Integration

1. Initialize 18-cell cable (8 atrial + 10 ventricular)
2. Stimulate SA node at `t=sa_fire_time`
3. Advance `cable.advance(dt)` every 33 ms
4. Cable returns dict of newly activated nodes
5. Accumulate activation times incrementally
6. Compute dipole moments from `activation_times`
7. Project to 12 leads

Architecture:

- `_cable`: Cable1D object (manages state V, W)
- `_active_beats`: List of BeatRecord objects
- `_current_beat_nodes`: Dict tracking activated nodes per beat
- `_nodes`: 18 ECGNode objects (anatomical positions)
- `_forward_model`: LeadField for projection

4 GUI Layer

4.1 PyQt6 Widget Hierarchy

```

MainWindow (QMainWindow)
    ControlPanel (QWidget)
        HR Spinner (QSpinBox)
        Engine Selector (QComboBox)
        Control Buttons (QPushButton)
        Display Label (QLabel)
    ECGDisplay (PlotWidget from pyqtgraph)
        12 curves (one per lead)
    ParameterPanel (QScrollArea)
        SANodeGroup (QGroupBox)
        AVNodeGroup (QGroupBox)
        ... (12 groups total)
        ApplyButton (QPushButton)

```

4.2 Signal/Slot Architecture

```

gray#gray grayUsergray graychangesgray grayHRgray grayspinbox
spinbox.valueChanged.connect(self.on_hr_changed)

bluedef on_hr_changed(self, new_hr):
    self.params.sa.cycle_length = 60 / new_hr
gray    gray#gray grayChangesgray graytakegray grayeffectgray grayongray
        graynextgray graybeat

gray#gray grayTimergray grayupdatesgray grayECGgray graydisplay
self.timer.timeout.connect(self.update_display)

bluedef update_display(self):
gray    gray#gray grayAdvancegray graysimulation
        self.engine.advance(DT_FRAME)

gray    gray#gray grayGetgray grayECGgray grayvoltages

```



```

        ecg_data = self.engine.get_ecg_data()

gray    gray#gray grayUpdategray grayallgray gray12gray graycurves
    bluefor i, lead bluein blueenumerate(self.leads):
        self.plot.getPlotItem().curves[i].setData(ecg_data[i])

```

5 Plugin System

5.1 Plugin Architecture

PluginBase (abstract)

StaticPlugin (cached, whole-beat effect)

StatefulPlugin (per-beat, temporary override)

Stacking Mechanism:

1. Phase 1/2 BFS algorithm runs
2. For each plugin in stack:
 - Modify activation_times dict or conductances
 - Can disable regions (set conductance $\rightarrow$ 0)
 - Can prolong refractories
3. Result: Composite arrhythmia

5.2 Available Plugins

Plugin	Mechanism
AVBlockI	Prolong av_delay
AVBlockIIMobitz1	Reduce av_conductance gradually
AVBlockIIMobitz2	Reduce his_conductance abruptly
AVBlockIII	Set av_conductance $\rightarrow$ 0
LBBB	Set lbb_conductance $\rightarrow$ 0
RBBB	Set rbb_conductance $\rightarrow$ 0
LAHB	Set lahb_conductance $\rightarrow$ 0
LPHB	Set lphb_conductance $\rightarrow$ 0
AF	Insert fibrillatory waves into atrial rate
PVC	Inject ectopic stimulus at coupling interval
Sinus Bradycardia	Increase sa_cycle_length

6 Data Flow Example: Normal Beat

1. User sets HR = 70 bpm
2. GUI calculates cycle_length = $60/70 \approx 0.857$ s
3. Timer tick at t=0: SA fires
4. BFS/Cable propagates: SA $\rightarrow$ RA (124 ms) $\rightarrow$ LA (191 ms) $\rightarrow$ AV (307 ms) $\rightarrow$ His (435 ms) $\rightarrow$ LV (645 ms)
5. Dipole computation for each region
6. Lead field projection: 18-D position vector $\rightarrow$ 12-lead voltages
7. Display updates all 12 curves
8. Next cycle_length : SA fires again

7 Performance Characteristics

7.1 Computational Complexity

Engine	BFS Complexity	Actual Rate	
Phase 1/2	$O(V + E) = O(18 + \text{edges})$	~ 1000 Hz	
Phase 3	$O(N_{cells}N_{steps})$	$2.2\times$ real-time	

7.2 Memory Usage

- State arrays: $18 \text{ cells} \times 2 \text{ variables (V, W)} \times 8 \text{ bytes} = 288 \text{ bytes}$
- ECG buffer: $10 \text{ seconds} \times 30 \text{ FPS} \times 12 \text{ leads} \times 8 \text{ bytes} = 28.8 \text{ KB}$
- Activation times: $18 \text{ nodes} \times 100 \text{ beats} \times 8 \text{ bytes} = 14.4 \text{ KB}$
- Total: ~ 100 MB for typical 1-hour simulation

8 Extension Points

8.1 Adding a New Cell Model

1. Create subclass of BaseCellModel
2. Implement: `get_initial_state()`, `compute_derivatives()`
3. Add to `cable_1d.py` imports
4. Update engine selector to include new model

8.2 Adding a New Plugin

1. Create class inheriting PluginBase
2. Implement: `apply()` method (modifies activation times or conductances)
3. Register in plugin manager
4. Add to GUI parameter panel

8.3 Adding a New Lead Vector

1. Define unit vector (X, Y, Z) in cardiac coordinate system
2. Add to LEAD_MATRIX in `lead_field.py`
3. Update lead names list
4. GUI automatically includes in display

ECG Simulator Documentation

Part 7: Technical Reference & Validation

Numerical Methods, CFL Stability, Performance Benchmarks

June 2026

Contents

1 CFL Stability Analysis

1.1 Finite-Difference Discretization

For the 1-D monodomain cable:

$$\frac{\partial v}{\partial t} = D \frac{\partial^2 v}{\partial x^2} + I_{\text{ion}}(v, w) \quad (1)$$

Explicit Euler in space (central difference) and time:

$$v_i^{n+1} = v_i^n + \Delta t \left[D \frac{v_{i+1}^n - 2v_i^n + v_{i-1}^n}{(\Delta x)^2} + I_{\text{ion}} \right] \quad (2)$$

1.2 CFL (Courant–Friedrichs–Lewy) Condition

For stability of explicit Euler:

$$\text{CFL} = \frac{D\Delta t}{(\Delta x)^2} \leq 0.5 \quad (3)$$

1.3 Validation for Phase 3 Cable

Region	D (cm ² /s)	Δx (cm)	CFL	Status	
Atrium (RA/LA)	0.288	0.2	0.192		
Purkinje (LBB)	7.2	0.2	0.48		
LV Myocardium	0.30	0.2	0.02		
Max CFL			0.48	Safe	

All regions satisfy CFL 0.5. Explicit Euler is stable for chosen parameters.

2 Numerical Integration Parameters

2.1 Phase 3 Cable Integration

Parameter	Value	Purpose	
τ_s (time scale)	0.015 s	Physical time normalization	
Δt_{int} (micro step)	0.0001 s	Integration step	
Δt_{frame} (display)	0.033 s	GUI update (30 FPS)	
N_{micro} (steps/frame)	$\text{ceil}(0.033 / 0.0001) = 330$	Steps per display update	

2.2 Phase 3-D Calibration

Numerical search for τ_s matching clinical ECG timings:

τ_s (ms)	P-wave (ms)	PR (ms)	QRS (ms)	
13	—	—	— (CFL violated)	
15	64	196	111 (SELECTED)	
20	104	266	151	
25	191	435	210	

Selected $\tau_s = 15$ ms matches all clinical targets.

3 Clinical Validation Data

3.1 Nominal ECG Parameters (Normal Sinus Rhythm)

Parameter	Simulated	Clinical Normal	Status	
HR	70 bpm	60–100 bpm		
P-wave duration	64 ms	40–100 ms		
PR interval	196 ms	120–200 ms		
QRS duration	111 ms	80–120 ms		
QT interval	420 ms	360–440 ms (male)		

3.2 Amplitude Validation

Wave	Simulated	Clinical Normal	Status	
QRS amplitude	1.287 mV	0.5–2.5 mV		
T-wave amplitude	0.590 mV	0.1–0.6 mV		
QRS/T ratio	2.18	1.5–3.0		

4 Performance Benchmarks

4.1 Computational Speed

Engine	Speed	Notes	
Phase 1/2 (BFS)	~1000 Hz	GPU-limited by display refresh	
Phase 3 (FHN Cable)	2.2× real-time	Continuous real-time capability	
Display refresh	30 FPS	PyQtGraph update rate	

4.2 Multi-Beat Performance

At 70 bpm (HR) with Phase 3 cable:

- **6 beats in 5 seconds:** Confirmed
- **Emergent refractoriness:** FHN recovery time $\tau_w \approx 234$ ms
- **Real-time display:** Smooth scrolling, no lag

5 Installation & Dependencies

5.1 System Requirements

- **OS:** Linux, macOS, or Windows
- **Python:** 3.10 or newer
- **RAM:** 512 MB minimum, 2 GB recommended
- **Disk:** 100 MB for source code + dependencies

5.2 Python Dependencies

Package	Version	Purpose	
PyQt6	6.0	GUI framework	
NumPy	2.0	Numerical computations	
pyqtgraph	0.12	Real-time ECG plotting	
scipy	1.10	Optional: advanced integration	

5.3 Installation Command

```
pip install PyQt6>=6.0 numpy>=2.0 pyqtgraph>=0.12
```

5.4 Running the Simulator

```
cd /path/to/ECG_simulator
python main.py
```

6 Glossary of Terms

Term	Definition
Action Potential (AP)	Rapid, temporary change in electrical potential of a cell
Automaticity	Ability of cardiac tissue to spontaneously depolarize
AV Node	Tissue between atria and ventricles; delays impulse
Bundle Branch	Part of conduction system in ventricles
CFL Condition	Courant–Friedrichs–Lewy stability criterion for explicit methods
Conduction Velocity	Speed at which electrical impulse travels through tissue
Depolarization	Change from resting potential to positive values
ECG/EKG	Electrocardiogram; electrical recording of heart
Ectopic Focus	Abnormal site of electrical activity
Excitability	Ease with which tissue can be stimulated
Fibrillation	Rapid, disorganized electrical activity
FitzHugh-Nagumo	Simplified 2-variable cell model
His Bundle	Part of conduction system; junction of AV node and bundle branches
Hyperpolarization	Potential more negative than resting (recovery phase)
Inactivation	State where ion channels cannot reopen immediately
Ischemia	Reduced blood flow to tissue
Lead Vector	Direction of electrical observation for 12-lead ECG
Monodomain	Mathematical model of continuous cardiac tissue
Refractoriness	Period when tissue cannot be stimulated
Repolarization	Return to resting potential
SA Node	Sinoatrial node; natural pacemaker of heart

7 References

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Part 8: Advanced Reference & Future Roadmap

Extended Features, Phase 4+ Planning, Troubleshooting

June 2026

Contents

1 Troubleshooting Guide

1.1 Flat QRS (No Visible QRS Complex)

1.1.1 Symptoms

- ECG shows flat or minimal voltage during QRS interval
- P waves visible (normal)
- T waves visible (normal)
- Only QRS appears depressed or absent

1.1.2 Root Causes

1. **Depol Gaussian decay:** QRS Gaussian has small σ_d (default 12.7 ms)
 - Solution: Increase depol_duration to 40 ms for Phase 3 (LV overlap)
2. **Amplitude scaling too low:** amplitude_scale $\neq$ 1.0
 - Solution: Set amplitude_scale = 1.0 or higher
3. **Engine selection wrong:** Using Phase 1/2 with Phase 3 parameters
 - Solution: Ensure engine selector matches code logic

1.1.3 Resolution Steps

1. Check GUI amplitude_scale slider (should be 1.0)
2. Verify Engine = "graph" (Phase 1/2) or "cable" (Phase 3)
3. For Phase 3: Ensure depol_duration parameter = 40 ms
4. Click Reset button to clear old traces
5. Click Start to generate new beat

1.2 No ECG Signal Generated

1.2.1 Symptoms

- Click Start button; no ECG trace appears
- Console shows no errors
- Display remains blank

1.2.2 Root Causes

1. **Timer not running:** GUI timer not started
2. **Parameters invalid:** Infinite or negative values
3. **Engine disconnected:** Engine not properly initialized

1.2.3 Resolution

1. Check console output (enable `debug_mode = true`)
2. Verify HR is between 30–220 bpm
3. Verify all conductances are 0.0–1.0
4. Reset GUI: Click Reset button
5. Try different engine (graph vs cable)

1.3 Irregular Rhythm When Expecting Regular

1.3.1 Causes

1. **AF enabled:** `af_enabled = true`
 - Solution: Set `af_enabled = false`
2. **Ectopic focus active:** `ectopic_enabled = true`
 - Solution: Set `ectopic_enabled = false`
3. **Escape pacemaker interfering:** `escape_enabled = true`
 - Solution: Ensure AV conductance = 1.0 so escapes don't fire

2 Advanced Parameter Tuning

2.1 Matching Patient-Specific ECG Patterns

To simulate a patient's ECG with specific morphology:

1. **Measure patient ECG:**
 - HR, PR, QRS, QT intervals
 - QRS axis, QRS amplitude
 - Presence of pathology (bundle block, ischemia, etc.)
2. **Adjust simulator parameters:**
 - Set HR to patient HR
 - Prolonged PR → increase `av_delay`
 - Broad QRS → reduce `lbb_conductance` or `rbb_conductance`
 - Left axis deviation → enable LAHB (set `lahb_conductance` → 0)
 - Right axis deviation → enable LPHB (set `lphb_conductance` → 0)
3. **Adjust timing constants (Phase 3):**
 - Modify τ_s in `cable_1d.py` to match patient's conduction times
 - Re-run numerical search for phase calibration
4. **Validate against patient ECG**
 - Compare intervals and amplitudes
 - Iterate parameter adjustments

3 Phase 4 Development Plan

3.1 Phase 4a: 3-D Ventricular Tissue (Q2 2027)

3.1.1 Objectives

- Extend Phase 3 cable model to 2-D ventricular sheet
- Implement orthotropic diffusion (longitudinal & transverse CV)
- Model 3-D dipole orientation changes during repolarization

3.1.2 Technical Changes

- Replace 1-D Laplacian with 2-D: $\nabla^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}$
- Introduce diffusion tensors: $D_L > D_T$ (anisotropy)
- Update CFL condition for 2-D: $\text{CFL} = D_{\max} \Delta t / \min((\Delta x)^2, (\Delta y)^2)$

3.1.3 Expected Outcomes

- Improved QRS morphology (especially V1–V3 precordial leads)
- Realistic T-wave vector changes
- Ability to simulate infarct patterns (regional conduction delay)

3.2 Phase 4b: Full Conduction System (Q3 2027)

3.2.1 Objectives

- Upgrade Phase 3 cable to full 3-D geometry
- Include:
 - SA node (pacemaker dynamics)
 - Complete atrial geometry
 - AV node (two-dimensional with slow-fast regions)
 - His bundle and bundle branches
 - Complete LV and RV

3.2.2 New Parameters

- Fast-pathway conductance (AV node)
- Slow-pathway conductance (dual AV node pathways)
- Accessory pathway (Wolff–Parkinson–White syndrome)

3.3 Phase 5: Advanced Cell Models (Q4 2027)

3.3.1 Courtemanche Model (Atrial)

- 21-variable state vector (vs 2 for FHN)
- Detailed ion channels: I_{Na} , I_K , I_{Ca} , I_{K1} , ...
- APD typically 100–150 ms (clinically accurate)
- Will enable: AF dynamics, APD alternans

3.3.2 ten Tusscher Model (Ventricular)

- 19-variable state vector
- Epicardial, midmyocardial, endocardial variants
- APD typically 200–300 ms
- Will enable: Transmural dispersion, long QT syndrome

3.4 Phase 6: Boundary-Element Forward Model (2028)

3.4.1 Replace Single-Dipole with BE Torso Model

- Construct 3-D torso geometry (from CT/MRI templates)
- Compute BEM gain matrix (heart $\rightarrow$ surface electrodes)
- Project 3-D ventricular dipoles to 12 standard + additional leads

3.4.2 Expected Improvements

- Realistic lead vectors (St.-axis angles match clinical)
- Ability to model unusual electrode placements
- ST-segment morphology improvements

3.5 Phase 7: Autonomic Nervous System (2028)

3.5.1 Sympathetic/Parasympathetic Modulation

- Model beta-adrenergic effects: increased HR, faster AV conduction
- Model parasympathetic effects: decreased HR, AV block risk
- Parameters: **sympathetic_tone**, **parasympathetic_tone**
- Realistic response to exercise, stress, sleep

3.5.2 Pharmacological Model

- Beta-blockers: Reduce HR, prolong PR
- Calcium channel blockers: AV nodal delay
- Antiarrhythmics: Class I (Na channel), Class II (beta), Class III (K channel)
- Simulate drug effects on conduction and refractoriness

4 Extending the Code

4.1 Adding a Custom Pathology Plugin

4.1.1 Step 1: Create Plugin Class

```

from plugins.base import PluginBase

class MyCustomArrhythmia(PluginBase):
    name = "MyCustomArrhythmia"
    description = "My custom ECG pattern"

    def __init__(self, params):
        self.params = params
        self.enabled = False

    def apply(self, activation_times, conductances):
        if not self.enabled:
            return activation_times, conductances

        # Modify activation times or conductances here
        # Example: block AV node
        conductances['av'] *= 0.5

        return activation_times, conductances

```

4.1.2 Step 2: Register Plugin

```

# In main.py or plugin manager
from plugins.my_custom import MyCustomArrhythmia

engine.register_plugin(MyCustomArrhythmia(params))

```

4.1.3 Step 3: Add GUI Controls

1. Add group box to parameter panel
2. Add enable/disable checkbox
3. Add parameter sliders (e.g., severity, timing)
4. Connect to plugin.apply() method

4.2 Modifying Conduction Geometry

To change the anatomical connectivity (DAG in Phase 1/2):

1. Edit cardiac_sim/core/tissue/graph.py
2. Modify adjacency matrix or edge list
3. Update conduction delays (path_delay dict)
4. Test with BFS algorithm

4.3 Adding New ECG Leads

1. Define new unit vector in cardiac coordinates
2. Add to LEAD_MATRIX in lead_field.py
3. Update lead names list
4. GUI automatically displays new lead

5 Known Limitations

1. **Single-dipole ECG model:** Limited accuracy for far-field effects
2. **No T-wave alternans:** Current model lacks beat-to-beat APD variation
3. **No ischemia simulation:** No regional metabolic effects on conduction
4. **Simplified atrial geometry:** 8-cell cable vs realistic 3-D atria
5. **No autonomic integration:** No HR variability from breathing, stress
6. **FHN cell model:** APD 540 ms (longer than clinical 200–300 ms for ventricle)

6 Quick Reference: Parameter Summary

Parameter	Range	Default	
HR (bpm)	30–220	70	
cycle_length (s)	0.27–2.0	0.857	
av_delay (s)	0.05–0.30	0.15	
av_conductance	0.0–1.0	1.0	
vent_abs_refr (s)	0.25–0.45	0.35	
amplitude_scale	0.1–5.0	1.0	

7 Contact & Support

For bug reports, feature requests, or documentation clarifications:

- **Repository:** [GitHub URL to ECG simulator]
- **Documentation:** See Part 1–7 for comprehensive reference
- **Issues:** File issue on GitHub with reproducible steps
- **Discussion:** Review Phase 4+ planning in this Part 8